Standard Model Signal in Cosmological Higgs Collider

Abstract

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1 Introduction

2 Gauge Boson Signal

2.1 Preliminaries

We will compute the connected four-point function $\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'$ generated by a massive gauge-boson loop. Our interest is the leading nonlocal signal in the collapsed s channel,

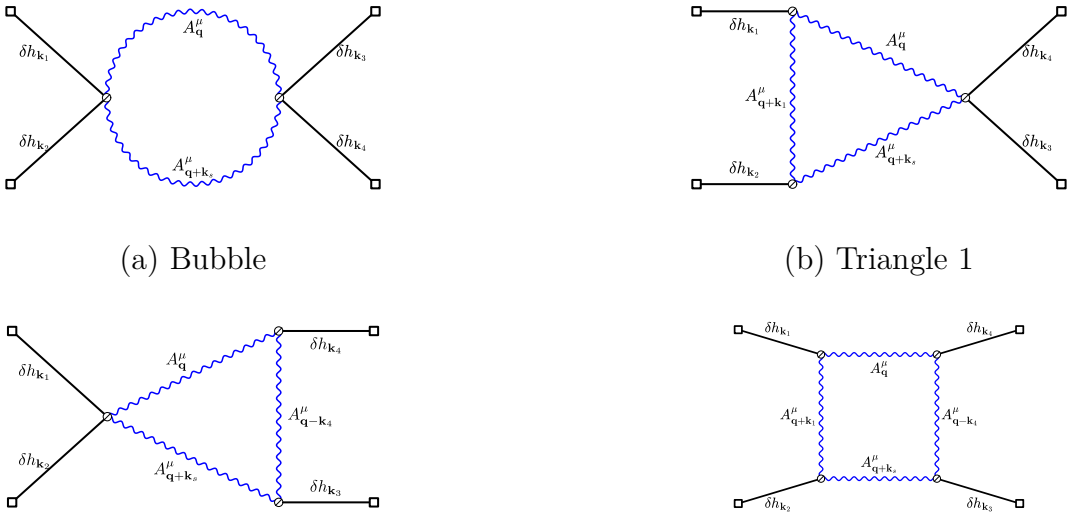
$$\mathbf{k}_s \equiv \mathbf{k}_1 + \mathbf{k}_2 = -(\mathbf{k}_3 + \mathbf{k}_4), \qquad k_s \ll k_i, \qquad k_{12} \equiv k_1 + k_2, \quad k_{34} \equiv k_3 + k_4, \tag{1}$$

where $k_i \equiv |\mathbf{k}_i|$. Thus $k_1 \simeq k_2$ and $k_3 \simeq k_4$ at leading order. We choose $\mathbf{k}_s$ along the z axis and set the azimuth of $\mathbf{k}_1$ to zero. Here θ_1 and θ_3 are the polar angles of $\mathbf{k}_1$ and $\mathbf{k}_3$, respectively, and ϕ_3 is their relative azimuth.

The Higgs–gauge couplings follow from the Higgs kinetic term [1]:

$$\frac{\sqrt{-g}}{2} |\mathrm{D}_\mu \mathbf{H}|^2 \supset \frac{a^2 g^2}{4} (h_0 + \delta h)^2 \left(W_\mu^+ W^{-\mu} + \frac{Z_\mu Z^\mu}{2 c_W^2} \right), \tag{2}$$

where g is the weak gauge coupling, $c_W = \cos\theta_W$, and the Lorentz contractions on the right use the flat conformal metric. The terms linear and quadratic in δh generate the cubic and quartic vertices, respectively. They give the bubble, two triangle placements, and box shown in Fig. 1.



(a) Bubble (b) Triangle 1

(c) Triangle 2 (d) Box

Figure 1: The four gauge-boson loop diagrams considered here. Black solid lines denote external Higgs modes and blue wavy lines denote gauge-boson propagators.

In the light-Higgs regime, the dominant infrared Higgs-loop contribution to the vector masses is [1]

$$m_W^2 \simeq \frac{3 g^2 H^4}{8\pi^2 m_H^2}, \qquad m_Z^2 \simeq \frac{m_W^2}{c_W^2}. \tag{3}$$

Here m_H is the effective Higgs mass. We use a constant vector mass m_V and the parameter $\tilde{\nu} \equiv \sqrt{m_V^2/H^2 - 1/4} > 0$ in the following. The W and Z contributions are obtained with their respective masses and couplings in (2).

We use $\lambda, \sigma = L, +, -, T$ for vector polarization-basis labels, and $\eta = \pm 1$ for the two nonlocal branches. Neglecting the Higgs mass in the external mode functions, we write its bulk-to-boundary propagator as

$$G_{\mathsf{a}}(k;\tau) = \frac{H^2}{2k^3}(1 - \mathrm{i}\mathsf{a}k\tau)e^{\mathrm{i}\mathsf{a}k\tau}. \tag{4}$$

The spacetime vector propagator decomposes as

$$\begin{aligned} D_{\mathsf{ab},\mu\nu}(\mathbf{k};\tau_1,\tau_2) = &\sum_{\lambda=T,L,\pm} e^{(\lambda)}_{\mathbf{k},\mu} e^{(\lambda)*}_{\mathbf{k},\nu} D^{(\lambda)}_{\mathsf{ab}}(k;\tau_1,\tau_2) \\ &+ e^{(T)}_{\mathbf{k},\mu} e^{(L)*}_{\mathbf{k},\nu} D^{(TL)}_{\mathsf{ab}}(k;\tau_1,\tau_2) + e^{(L)}_{\mathbf{k},\mu} e^{(T)*}_{\mathbf{k},\nu} D^{(LT)}_{\mathsf{ab}}(k;\tau_1,\tau_2), \end{aligned} \tag{5}$$

where $\mu, \nu = 0, 1, 2, 3$. In the comoving frame, the covariant polarization basis is [2]

$$e^{(T)}_{\mathbf{k},\mu} = (1, \mathbf{0}), \qquad e^{(L)}_{\mathbf{k},\mu} = (0, \widehat{\mathbf{k}}), \qquad e^{(\pm)}_{\mathbf{k},\mu} = \frac{1}{\sqrt{2}}(0, \mathbf{e}_\theta \pm \mathrm{i}\mathbf{e}_\phi),$$

with $(\widehat{\mathbf{k}}, \mathbf{e}_\theta, \mathbf{e}_\phi)$ the orthonormal spherical basis. The label T denotes the temporal component of the longitudinal mode.

At zero chemical potential, all components can be expressed through an auxiliary scalar propagator with the same Hankel index [2, 3]:

$$D^{(\mathrm{s})}_{>}(k;\tau_1,\tau_2) = \frac{\pi H^2}{4} e^{-\pi\tilde{\nu}} (\tau_1\tau_2)^{3/2} \mathrm{H}^{(1)}_{\mathrm{i}\tilde{\nu}}(-k\tau_1) \mathrm{H}^{(2)}_{-\mathrm{i}\tilde{\nu}}(-k\tau_2), \tag{6}$$

$$D^{(\pm)}_{>} = \frac{D^{(\mathrm{s})}_{>}}{H^2\tau_1\tau_2}, \qquad D^{(L)}_{>} = \frac{(\partial_{\tau_1} - 2/\tau_1)(\partial_{\tau_2} - 2/\tau_2) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \tag{7}$$

$$D^{(T)}_{>} = \frac{k^2 D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \tag{8}$$

$$D^{(TL)}_{>} = \frac{\mathrm{i}k(\partial_{\tau_2} - 2/\tau_2) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}, \qquad D^{(LT)}_{>} = -\frac{\mathrm{i}k(\partial_{\tau_1} - 2/\tau_1) D^{(\mathrm{s})}_{>}}{H^2(\tilde{\nu}^2 + 1/4)}. \tag{9}$$

With spacetime indices and momentum arguments suppressed, $D_{-+} = D_>$, $D_{+-} = D_<$, and the separated-time SK propagators are

$$\begin{aligned} D_{++} &= \theta(\tau_1 - \tau_2) D_> + \theta(\tau_2 - \tau_1) D_<, \\ D_{--} &= \theta(\tau_2 - \tau_1) D_> + \theta(\tau_1 - \tau_2) D_<. \end{aligned} \tag{10}$$

These relations apply to $D^{(\pm)}, D^{(L)}, D^{(TL)}$ and $D^{(LT)}$. But for $D^{(T)}$, the $++$ and $--$ propagators have a delta-function term because the temporal mode is not a physical degree of freedom:

$$\begin{aligned} D^{(T)}_{++} &= \theta(\tau_1 - \tau_2) D^{(T)}_{>} + \theta(\tau_2 - \tau_1) D^{(T)}_{<} + \frac{\mathrm{i}\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4) a(\tau_1)^2}, \\ D^{(T)}_{--} &= \theta(\tau_2 - \tau_1) D^{(T)}_{>} + \theta(\tau_1 - \tau_2) D^{(T)}_{<} - \frac{\mathrm{i}\delta(\tau_1 - \tau_2)}{H^2(\tilde{\nu}^2 + 1/4) a(\tau_1)^2}. \end{aligned} \tag{11}$$

We will see that the propagators $D^{(T)}, D^{(TL)}$ and $D^{(LT)}$ will not contribute to the leading nonlocal signal.

The leading nonlocal parts of the spatial propagators are independent of the SK indices:

$$\left[D^{(\pm)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{(\tau_1\tau_2)^{1/2}}{4\pi}\sum_{\eta=\pm1}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}, \tag{12}$$

$$\left[D^{(L)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{(\tau_1\tau_2)^{1/2}}{4\pi}\sum_{\eta=\pm1}\frac{(-1/2+\mathrm{i}\eta\tilde{\nu})^2}{\tilde{\nu}^2+1/4}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}. \tag{13}$$

The temporal nonlocal part is

$$\left[D^{(T)}_{\mathsf{ab}}(k;\tau_1,\tau_2)\right]_{\rm NonL} = \frac{k^2(\tau_1\tau_2)^{3/2}}{4\pi(\tilde{\nu}^2+1/4)}\sum_{\eta=\pm1}\Gamma^2(-\mathrm{i}\eta\tilde{\nu})\left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}}. \tag{14}$$

The one-loop factorization theorem and cutting rule [4] isolate the leading nonlocal signal in the collapsed channel. We first identify the two internal lines and their momenta can be chosen as $\mathbf{q}$ and $\mathbf{K}\equiv\mathbf{q}+\mathbf{k}_s$. The relevant loop region has $q, K\sim k_s \ll k_i$. We replace these two propagators by the leading nonlocal terms of their late-time expansions, keeping their polarization tensors, and expand the remaining hard propagators to leading order in the soft momenta. The hard propagators retain their full time dependence and SK ordering. The cut propagators are independent of the SK indices, and their time dependence separates between the two endpoints. The time integrals thus factorize into two tree subgraphs with additional time powers supplied by the cut lines. Multiplying these subgraphs by the soft momentum integral and summing over polarizations gives the nonlocal signal. The branches $\eta=\pm1$ combine into the oscillatory signal retained below. Higher terms in the soft expansion give subleading corrections in the collapsed limit.

Now we will show that the propagators $D^{(T)}$, $D^{(TL)}$, and $D^{(LT)}$ do not contribute to the leading nonlocal signal. On a soft line, each temporal mode function $B^{(T)}$ carries an additional factor of $k\tau$ relative to the spatial modes $B^{(L)}$ and $B^{(\pm)}$. Their nonanalytic contributions are therefore suppressed in the collapsed limit. The contact term in $D^{(T)}$ is analytic in the soft momentum and does not contribute to the nonlocal signal. On a hard line, a temporal endpoint has a vanishing polarization contraction with the spatial components of the attached soft line. A nonzero contraction would require a temporal soft component, which is already suppressed. We therefore omit propagators with temporal components when computing the leading signal below.

All required soft momentum integrals reduce to five dimensionless functions f_r, $r=0,1,2,3,4$. With $z\equiv\widehat{\mathbf{q}}\cdot\widehat{\mathbf{k}}_s$, we define

$$\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}q^{-2\alpha}K^{-2\beta}z^r = \frac{k_s^{3-2\alpha-2\beta}}{(4\pi)^{3/2}}f_r(\alpha,\beta), \qquad r=0,1,2,3,4. \tag{15}$$

These expressions are understood by analytic continuation from the convergent domain. The first three functions are

$$f_0(\alpha,\beta) = \frac{\Gamma(\alpha+\beta-3/2)\Gamma(3/2-\alpha)\Gamma(3/2-\beta)}{\Gamma(3-\alpha-\beta)\Gamma(\alpha)\Gamma(\beta)}, \tag{16}$$

$$f_1(\alpha,\beta) = \frac{1}{2}\left[f_0(\alpha+\tfrac{1}{2},\beta-1) - f_0(\alpha-\tfrac{1}{2},\beta) - f_0(\alpha+\tfrac{1}{2},\beta)\right], \tag{17}$$

$$\begin{aligned} f_2(\alpha,\beta) = \frac{1}{4}\big[&f_0(\alpha+1,\beta-2) + f_0(\alpha-1,\beta) + f_0(\alpha+1,\beta) \\ &-2f_0(\alpha,\beta-1) - 2f_0(\alpha+1,\beta-1) + 2f_0(\alpha,\beta)\big]. \end{aligned} \tag{18}$$

The functions f_3 and f_4 are likewise finite sums of f_0 and their explicit expressions are collected in App. A.1.

For the nonlocal branch η, it is convenient to set $x = -\mathrm{i}\eta\tilde{\nu}$ and define

$$\mathcal{J}_r(m,n;x) \equiv f_r\left(x-\frac{m}{2}, x+\frac{n}{2}\right), \qquad r=0,1,2,3,4. \tag{19}$$

Here m, n are integer shifts of momentum powers, unrelated to m_V. In particular,

$$\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\left(\frac{q}{2}\right)^{-2x}\left(\frac{K}{2}\right)^{-2x} q^m K^{-n} z^r = \frac{4^{2x}k_s^{3-4x+m-n}}{(4\pi)^{3/2}}\mathcal{J}_r(m,n;x). \tag{20}$$

2.2 Bubble Diagram

The bubble contains two quartic vertices, each attached to a pair of external Higgs modes. With the four external momenta understood, the s-channel contribution is

$$\begin{aligned}\langle\delta h^4\rangle'_{\mathrm{Bub}} = \frac{g^4}{8}\sum_{\mathsf{a},\mathsf{b}=\pm1}(-\mathsf{ab})\int_{-\infty}^{0}\frac{\mathrm{d}\tau_1}{|H\tau_1|^2}\int_{-\infty}^{0}\frac{\mathrm{d}\tau_2}{|H\tau_2|^2}G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{a}}(k_2;\tau_1)G_{\mathsf{b}}(k_3;\tau_2)G_{\mathsf{b}}(k_4;\tau_2)\\ \times\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q};\tau_1,\tau_2)D_{\mathsf{ba},ji}(\mathbf{K};\tau_2,\tau_1), \qquad \mathbf{K}=\mathbf{q}+\mathbf{k}_s.\end{aligned} \tag{21}$$

The symmetry factor $1/2$ which is associated with exchanging the two identical internal lines is included. For the fixed partition (12) | (34), exchanges within either external Higgs pair leave the graph unchanged. There is therefore no additional external-permutation factor in the s-channel signal. And throughout this section, the $W^{\pm}$ contribution is obtained by multiplying the expressions by 2, and the Z contribution by making the replacement $g \to g/c_W$, with the corresponding mass used in each case.

Replacing the two soft propagators by (12)–(13) removes their SK-index dependence, as illustrated in Fig. 2. The oscillatory signal comes from the two soft propagators with the same branch label η. After summing the polarizations, this contribution is

$$\begin{aligned}&\left[\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q};\tau_1,\tau_2)D_{\mathsf{ba},ji}(\mathbf{K};\tau_2,\tau_1)\right]_{\mathrm{NonL}}\\ &\qquad\simeq\sum_{\eta=\pm1}(\tau_1\tau_2)^{1+2\mathrm{i}\eta\tilde{\nu}}\frac{\Gamma^4(-\mathrm{i}\eta\tilde{\nu})}{(4\pi)^{7/2}}4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}\mathcal{I}_{\mathrm{Bub}}(-\mathrm{i}\eta\tilde{\nu}).\end{aligned} \tag{22}$$

The dimensionless polarization sum has the compact form

$$\mathcal{I}_{\mathrm{Bub}}(x) = \frac{3+2x}{1-2x}\mathcal{J}_0(0,0;x) + \frac{16x^2}{(1-2x)^2}\left[\mathcal{J}_0(2,2;x)+2\mathcal{J}_1(1,2;x)+\mathcal{J}_2(0,2;x)\right], \tag{23}$$

with $x = -\mathrm{i}\eta\tilde{\nu}$ as in (19). The separate polarization contributions are given in App. A.2.

The remaining time integrals factorize. For the left subgraph, at $k_1 \simeq k_2 \simeq k_{12}/2$, we find

$$\sum_{\mathsf{a}=\pm1}\mathsf{i}\mathsf{a}\int_{-\infty}^{0}\frac{\mathrm{d}\tau}{|H\tau|^2}G_{\mathsf{a}}(k_1;\tau)G_{\mathsf{a}}(k_2;\tau)(-\tau)^{1+2\mathrm{i}\eta\tilde{\nu}} \simeq \frac{H^2k_{12}^{-2\mathrm{i}\eta\tilde{\nu}}}{4k_1^3k_2^3}\mathcal{P}_\eta^{(1)}(\tilde{\nu}), \tag{24}$$

$$\mathcal{P}_\eta^{(1)}(\tilde{\nu}) = \frac{2+\mathrm{i}\eta\tilde{\nu}}{2\mathrm{i}\eta\tilde{\nu}}\mathrm{i}\sinh(\pi\eta\tilde{\nu})\Gamma(2+2\mathrm{i}\eta\tilde{\nu}). \tag{25}$$

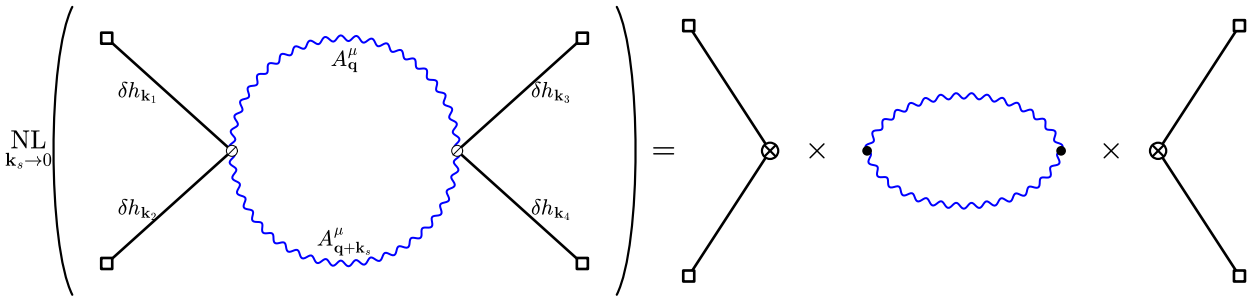


Figure 2: The leading nonlocal part of the bubble in the $k_s \to 0$ limit. The two soft gauge-boson lines supply the central loop factor, while the left and right subgraphs give the two quartic-vertex time integrals.

The right vertex gives the same expression with $(k_1, k_2, k_{12}) \to (k_3, k_4, k_{34})$. Combining the two vertices with (22), we obtain

$$\left[\langle \delta h^4 \rangle'_{\rm Bub}\right]_{\rm NonL} \simeq \frac{g^4 H^4 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2i\tilde{\nu}} C^{\rm Bub}(\tilde{\nu}) + \text{c.c.}\right], \tag{26}$$

where the complex signal coefficient is

$$C^{\rm Bub}(\tilde{\nu}) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \left[\mathcal{P}^{(1)}_{+}(\tilde{\nu})\right]^2 \mathcal{I}_{\rm Bub}(-i\tilde{\nu}). \tag{27}$$

This definition separates the loop and time integrals from the couplings and diagram symmetry factor. The bracket in (26) equals

$$2|C^{\rm Bub}(\tilde{\nu})| \cos\left[2\tilde{\nu}\log\left(\frac{k_s^2}{4k_{12}k_{34}}\right) + \arg C^{\rm Bub}(\tilde{\nu})\right]. \tag{28}$$

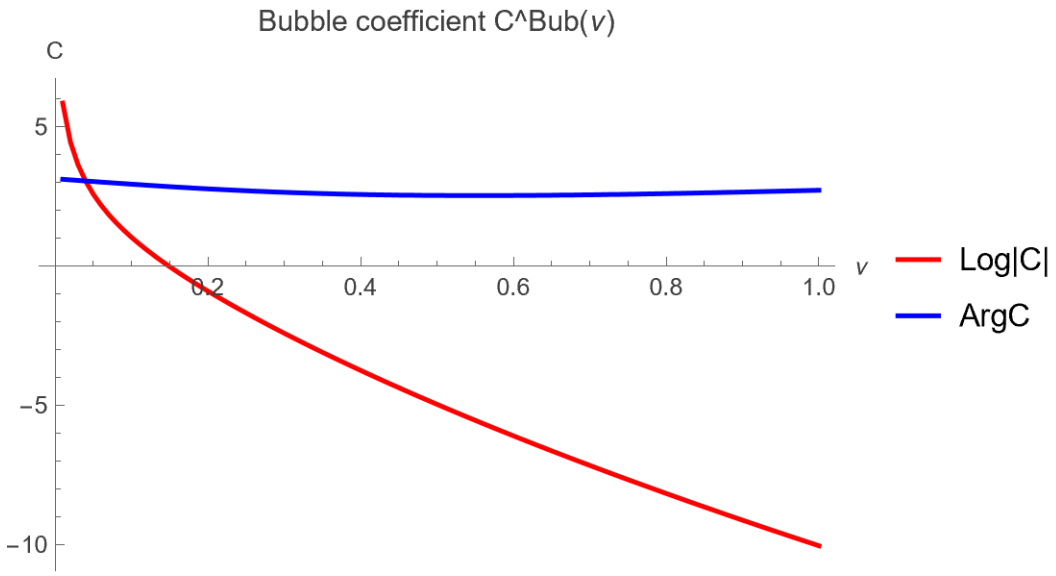


Figure 3: The bubble coefficient defined in (27). The red and blue curves show $\log|C^{\rm Bub}|$ and $\arg C^{\rm Bub}$ in radians, respectively, as functions of $\tilde{\nu}$.

2.3 Triangle Diagram

The triangle contains two cubic vertices and one quartic vertex. We first consider the diagram "Triangle 1". The triangle has a unit diagram symmetry factor. In the conventions of Sec. 2.2, its integral expression is

$$\langle\delta h^4\rangle'_{\rm Tri1} = \frac{g^6h_0^2}{8}\sum_{\mathsf{a},\mathsf{b},\mathsf{c}=\pm1}(-\mathrm{i}\mathsf{abc})\int_{-\infty}^{0}\prod_{v=1}^{3}\frac{\mathrm{d}\tau_v}{|H\tau_v|^2}\,G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{b}}(k_2;\tau_2)G_{\mathsf{c}}(k_3;\tau_3)G_{\mathsf{c}}(k_4;\tau_3)$$
$$\times\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q}+\mathbf{k}_1;\tau_1,\tau_2)D_{\mathsf{bc},j\ell}(\mathbf{K};\tau_2,\tau_3)D_{\mathsf{ca},\ell i}(\mathbf{q};\tau_3,\tau_1). \tag{29}$$

The nonlocal signal comes from $q, K \sim k_s \ll k_1$. We cut the $\mathbf{q}$ and $\mathbf{K}$ lines and evaluate the remaining hard propagator at $\mathbf{q}+\mathbf{k}_1 \simeq \mathbf{k}_1$. The latter retains its full time dependence and SK ordering. The resulting factorization is shown in Fig. 4.

After summing the soft polarizations, the cut loop takes the form

$$\left[\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}D_{\mathsf{ab},ij}(\mathbf{q}+\mathbf{k}_1;\tau_1,\tau_2)D_{\mathsf{bc},j\ell}(\mathbf{K};\tau_2,\tau_3)D_{\mathsf{ca},\ell i}(\mathbf{q};\tau_3,\tau_1)\right]_{\rm NonL}$$
$$\simeq\sum_{\lambda=L,\pm}\sum_{\eta=\pm1}D^{(\lambda)}_{\mathsf{ab}}(k_1;\tau_1,\tau_2)(\tau_1\tau_2)^{1/2+\mathrm{i}\eta\tilde{\nu}}(-\tau_3)^{1+2\mathrm{i}\eta\tilde{\nu}}\frac{\Gamma^4(-\mathrm{i}\eta\tilde{\nu})}{(4\pi)^{7/2}}4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}\mathcal{I}^{\lambda}_{\rm Tri}(-\mathrm{i}\eta\tilde{\nu},\theta_1). \tag{30}$$

This equation defines the dimensionless coefficients $\mathcal{I}^{\lambda}_{\rm Tri}(x,\theta)$, with $x=-\mathrm{i}\eta\tilde{\nu}$. After simplification, we surprisingly find that all three coefficients equal $\mathcal{I}_{\rm Bub}(x)/3$ and the apparent angular dependence cancels, as shown in (74).

We next evaluate the left and right subgraphs. The quartic vertex gives the time integral $\mathcal{P}^{(1)}_{\eta}$

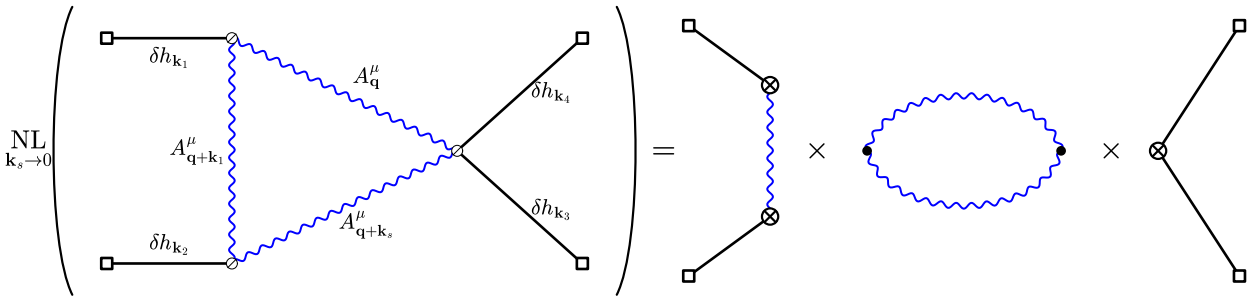


Figure 4: The leading nonlocal part of the first triangle. Cutting the two soft lines leaves a hard gauge-boson exchange on the left and a quartic vertex on the right, joined by the central soft loop factor. The second triangle follows by exchanging the two external pairs.

in (25). The two cubic vertices and the hard line instead define

$$\sum_{\mathsf{a},\mathsf{b}=\pm1}(-\mathsf{ab})\int_{-\infty}^{0}\frac{\mathrm{d}\tau_1\mathrm{d}\tau_2}{|H\tau_1|^2|H\tau_2|^2}G_{\mathsf{a}}(k_1;\tau_1)G_{\mathsf{b}}(k_2;\tau_2)D^{(\lambda)}_{\mathsf{ab}}(k_1;\tau_1,\tau_2)(\tau_1\tau_2)^{1/2+\mathrm{i}\eta\tilde{\nu}}$$
$$\simeq \frac{k_{12}^{-2\mathrm{i}\eta\tilde{\nu}}}{4k_1^3k_2^3}\mathcal{P}^{(2),\lambda}_{\eta}(\tilde{\nu}). \tag{31}$$

At leading order, $k_1 \simeq k_2 \simeq k_{12}/2$. The integral therefore reduces to the double-folded scalar seeds of Ref. [3], which we denote by $\mathcal{M}^{p_1p_2}_{\mathsf{ab}} \equiv \widetilde{\mathcal{I}}^{p_1p_2}_{\mathsf{ab}}(1,1)$. For the transverse modes this follows directly from (7). For the longitudinal mode, the differential relation also produces a contact contribution, and we retain this term together with the scalar-seed contributions. The required seed exponents include $p_0 = -5/2 + \mathrm{i}\eta\tilde{\nu}$, on the boundary of the convergence domain. We define the result by analytic continuation from $p_0 + \epsilon$, $\epsilon > 0$, and take the finite part of the assembled expression at $\epsilon = 0$. The seed normalization, closed forms, and combinations giving $\mathcal{P}^{(2),\lambda}_{\eta}$ are collected in App. A.3.

Finally, the leading signal of the first triangle is

$$\left[\langle\delta h^4\rangle'_{\mathrm{Tri1}}\right]_{\mathrm{NonL}} \simeq \frac{g^6h_0^2H^2k_s^3}{128k_1^3k_2^3k_3^3k_4^3}\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}}C^{\mathrm{Tri}}(\tilde{\nu}) + \text{c.c.}\right], \tag{32}$$

where

$$C^{\mathrm{Tri}}(\tilde{\nu}) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\mathcal{P}^{(1)}_{+}(\tilde{\nu})\sum_{\lambda=L,\pm}\mathcal{P}^{(2),\lambda}_{+}(\tilde{\nu})\mathcal{I}^{\lambda}_{\mathrm{Tri}}(-\mathrm{i}\tilde{\nu},\theta). \tag{33}$$

Since $\mathcal{P}^{(2),+}_{\eta} = \mathcal{P}^{(2),-}_{\eta}$, the two transverse helicities contribute equally. The simplified loop coefficients make C^{Tri} independent of θ at this order and its reduced expression is given in (75).

In the collapsed s channel, the quartic vertex carries either the pair $(3,4)$, as in Triangle 1, or the pair $(1,2)$, as in Triangle 2. For either placement, exchanging the two Higgs legs at the cubic vertices reproduces the same undirected triangle after reflection. Exchanging the two legs at the quartic vertex also leaves the graph unchanged, since both attach to the same vertex. Each placement is therefore counted once. Triangle 2 follows from Triangle 1 by $(1,2) \leftrightarrow (3,4)$, and

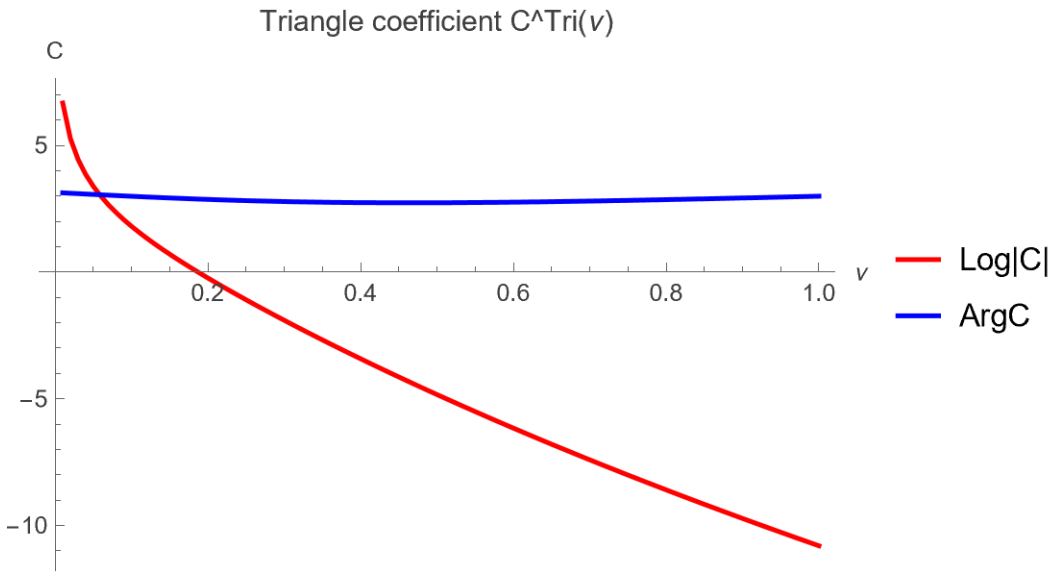


Figure 5: The coefficient $C^{\rm Tri}(\tilde{\nu})$ defined in (33). The red and blue curves show $\log|C^{\rm Tri}|$ and $\arg C^{\rm Tri}$ in radians, respectively.

their leading collapsed signals are equal. Adding these two contributions gives

$$\left[\langle\delta h^4\rangle'_{\rm Tri}\right]_{\rm NonL} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3}\left\{\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2i\tilde{\nu}}\left[C^{\rm Tri}(\tilde{\nu})+C^{\rm Tri}(\tilde{\nu})\right]+\text{c.c.}\right\}. \tag{34}$$

The nonanalytic momentum dependence is the same as in the bubble, while the hard propagator changes the amplitude and phase. Figure 5 shows the coefficient of a single triangle.

2.4 Box Diagram

We first consider the box with cyclic ordering 1234, shown in Fig. 6. With the external labels fixed, its diagram symmetry factor is unity. The four cubic vertices give the coupling factor $(g^2h_0/2)^4 = g^8h_0^4/16$.

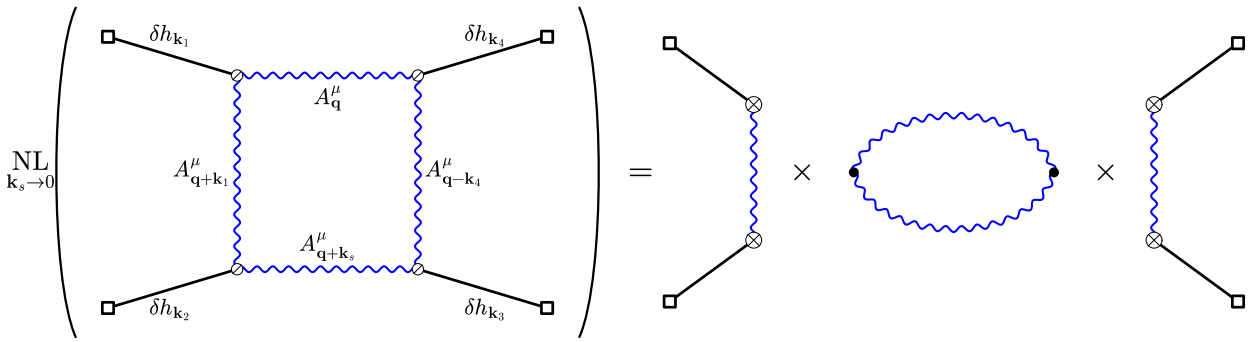


Figure 6: The leading nonlocal part of the box with cyclic ordering 1234. Cutting the two soft lines separates two tree subgraphs, each containing one hard gauge-boson propagator. The central factor contains the nonlocal parts of the two soft propagators and their loop momentum integral.

Choose the loop momenta along the four successive internal lines as $\mathbf{q}+\mathbf{k}_1$, $\mathbf{K} = \mathbf{q}+\mathbf{k}_s$, $\mathbf{q}-\mathbf{k}_4$, and $\mathbf{q}$. In the collapsed limit, the leading nonlocal signal comes from the region $q, K \sim k_s$. Cutting these two soft lines leaves two hard propagators, with momenta approaching $\mathbf{k}_1$ and $\mathbf{k}_3$. Their polarizations are denoted by $\lambda_1, \lambda_3 = L, \pm$. The soft loop gives the dimensionless coefficient $\mathcal{I}^{\lambda_1\lambda_3}_{\rm Box}(-\mathrm{i}\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3)$ defined in (81). Its common momentum factor is $4^{-2\mathrm{i}\eta\tilde{\nu}}k_s^{3+4\mathrm{i}\eta\tilde{\nu}}$, as for the bubble and triangle. As shown in Fig. 6, the time integrals factorize into two copies of (31), one for each hard line. They contribute $\mathcal{P}^{(2),\lambda_1}_\eta(\tilde{\nu})$ and $\mathcal{P}^{(2),\lambda_3}_\eta(\tilde{\nu})$.

The collapsed s channel contains a second inequivalent box, obtained from the diagram shown in Fig. 6 by exchanging the external momenta $\mathbf{k}_1$ and $\mathbf{k}_2$. Exchanging $\mathbf{k}_3$ and $\mathbf{k}_4$ gives the same additional undirected graph, while exchanging both pairs returns the original graph. At leading collapsed order, $\mathbf{k}_2 \simeq -\mathbf{k}_1$, so the first exchange sends (θ_1, ϕ_3) to $(\pi-\theta_1, \phi_3+\pi)$. The coefficient $C^{\rm Box}$ is invariant under this transformation, as follows from its angular decomposition below. The two boxes therefore contribute equally.

For each box, the vertex factor $1/16$ and the normalization $1/4$ from each hard subgraph give $1/256$. Summing the two distinct boxes supplies a further factor 2, giving $1/128$. Including the complex-conjugate soft branch, the total s-channel signal is

$$\left[\langle\delta h^4\rangle'_{\rm Box}\right]_{\rm NonL} \simeq \frac{g^8h_0^4k_s^3}{128k_1^3k_2^3k_3^3k_4^3}\left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}} \times C^{\rm Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.}\right], \tag{35}$$

Here $C^{\rm Box}$ denotes the coefficient of the representative box, before summing the external orderings:

$$C^{\rm Box}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}} \sum_{\lambda_1,\lambda_3=L,\pm} \mathcal{P}^{(2),\lambda_1}_+(\tilde{\nu})\mathcal{P}^{(2),\lambda_3}_+(\tilde{\nu}) \times \mathcal{I}^{\lambda_1\lambda_3}_{\rm Box}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3). \tag{36}$$

Since the hard transverse propagators have equal time integrals, the polarization sum reduces to the LL, LH, and HH groups in (82) where H means $+$ component contributions plus $-$ component contributions. Their explicit expressions are given in App. A.4.

The result contains five angular structures:

$$\begin{aligned} C^{\mathrm{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = {} & A_{1,g}(\tilde{\nu}) + A_{2,g}(\tilde{\nu})(\cos^2\theta_1 + \cos^2\theta_3) + A_{3,g}(\tilde{\nu})\cos^2\theta_1\cos^2\theta_3 \\ & + B_g(\tilde{\nu})\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 \\ & + D_g(\tilde{\nu})\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3). \end{aligned} \tag{37}$$

The five complex functions depend only on $\tilde{\nu}$. The signal is symmetric under $\theta_1 \leftrightarrow \theta_3$ and $\phi_3 \to -\phi_3$. Its oscillatory momentum dependence is the same as in (28), with an angle-dependent amplitude and phase supplied by C^{Box}.

Figures 7 and 8 show the five coefficient functions. Their magnitudes decrease over the displayed mass range. The jumps in the plotted phases reflect the principal branch of arg, defined modulo 2π.

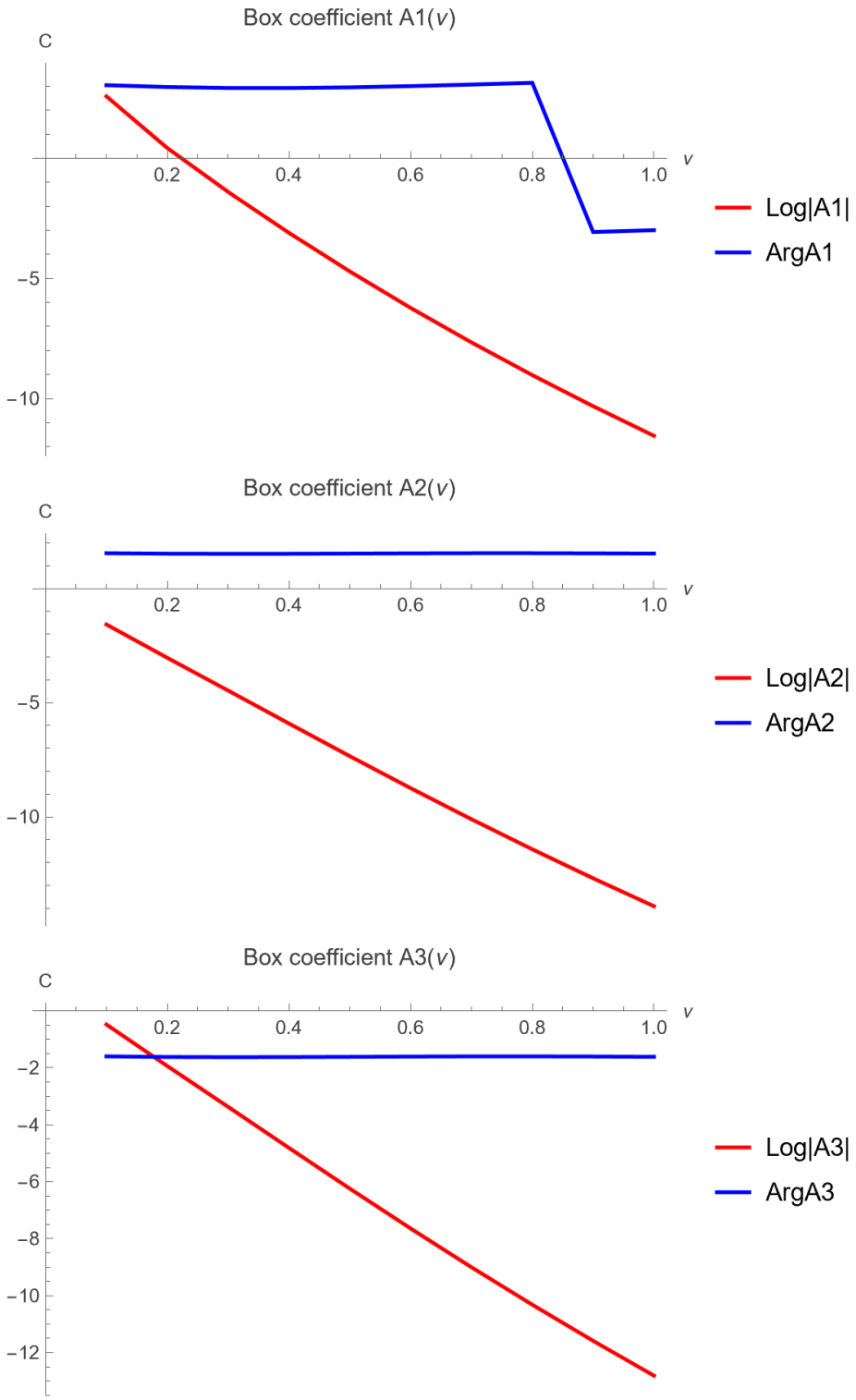


Figure 7: The angular coefficients $A_{1,g}$, $A_{2,g}$, and $A_{3,g}$ (top to bottom) in (37) for the box diagram. Red curves show the logarithmic magnitudes and blue curves show the principal phases in radians.

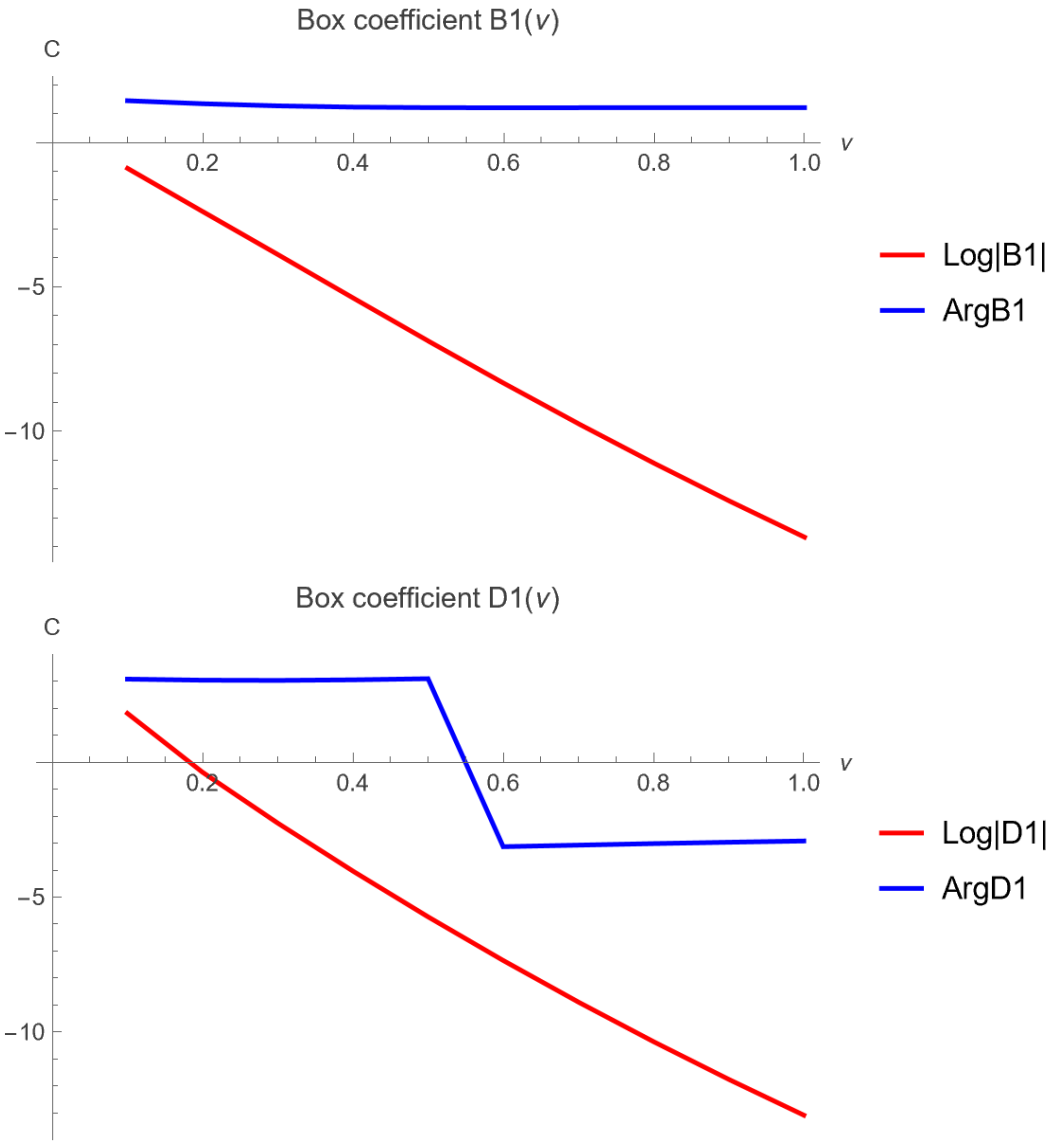


Figure 8: The angular coefficients B_g (top, labelled $B1$) and D_g (bottom, labelled $D1$) in (37). The plotting conventions are the same as in Fig. 7.

3 Top Quark Signal

3.1 Preliminaries

We next compute the Higgs four-point function generated by the top-quark loop. We use the collapsed kinematics of (1) and the external propagator (4). The top mass is treated as constant and the chemical potential vanishes. The Yukawa coupling and mass parameter are

$$g_t \equiv \frac{y_t}{\sqrt{2}}, \qquad m_t = g_t h_0, \qquad \tilde{\nu} \equiv \frac{m_t}{H} > 0. \tag{38}$$

Thus the relation between $\tilde{\nu}$ and the mass differs from that of the vector field.

Let t and t^c denote the conformally rescaled left-handed top and the left-handed charge conjugate of the right-handed top. Rescaling the covariant fields by $a^{3/2}$, with $a = -1/(H\tau)$, gives

$$\mathcal{L} = \mathrm{i}t^\dagger \bar{\sigma}^\mu \partial_\mu t + \mathrm{i}t^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(m_t + g_t \delta h)(t t^c + \mathrm{h.c.}). \tag{39}$$

Here t^c represents the right-handed degree of freedom in left-handed notation. Following the two-component description of Ref. [5], we set $t = (\psi_1 + \mathrm{i}\psi_2)/\sqrt{2}$ and $t^c = (\psi_1 - \mathrm{i}\psi_2)/\sqrt{2}$. The action becomes

$$\mathcal{L} = \sum_{I=1}^{2} \left[\mathrm{i}\psi_I^\dagger \bar{\sigma}^\mu \partial_\mu \psi_I - \frac{a}{2}(m_t + g_t \delta h)(\psi_I \psi_I + \mathrm{h.c.}) \right]. \tag{40}$$

The two sectors are identical. Each vertex retains one factor of $a(\tau)$ and the propagators below are those of the rescaled field and contain no factors of $a^{-3/2}$.

For one sector, the normalized Bunch–Davies state mode functions can be written as

$$u_+ = v_- = \frac{\tilde{\nu}}{\sqrt{2x}} W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad u_- = v_+ = \frac{\mathrm{i}}{\sqrt{2x}} W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad x = -k\tau. \tag{41}$$

The helicity is $h = \pm 1$, $|u_s|^2 + |v_s|^2 = 1$, and the creation part of the field contains v_h^*. We use the three propagators [6]

$$\begin{aligned} D_{\mathsf{ab},\alpha\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(\mathbf{k},\tau_1) \psi^\dagger_{\mathsf{b},\dot{\beta}}(-\mathbf{k},\tau_2)\rangle', \\ \widehat{D}_{\mathsf{ab},\alpha\beta}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(\mathbf{k},\tau_1) \psi_{\mathsf{b},\beta}(-\mathbf{k},\tau_2)\rangle', \\ \check{D}_{\mathsf{ab},\dot{\alpha}\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi^\dagger_{\mathsf{a},\dot{\alpha}}(\mathbf{k},\tau_1) \psi^\dagger_{\mathsf{b},\dot{\beta}}(-\mathbf{k},\tau_2)\rangle'. \end{aligned} \tag{42}$$

Here T_C denotes ordering along the Schwinger–Keldysh closed time contour C. Their mode sums and fermionic reversal rules are given in App. B.1. For each of the three types, the time-ordered components obey

$$\mathcal{D}_{++} = \theta_{12}\mathcal{D}_{-+} + \theta_{21}\mathcal{D}_{+-}, \qquad \mathcal{D}_{--} = \theta_{21}\mathcal{D}_{-+} + \theta_{12}\mathcal{D}_{+-}, \qquad \theta_{ij} = \theta(\tau_i - \tau_j). \tag{43}$$

To display the soft expansion, write $\sigma_p = \boldsymbol{\sigma}\cdot\hat{\mathbf{p}}$ and $\boldsymbol{\epsilon} = \mathrm{i}\sigma^2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $\epsilon_{12} = 1$ and $\epsilon^{12} = -1$. Define the common branch coefficient

$$c_\eta = \frac{2^{-1-2\mathrm{i}\eta\tilde{\nu}}}{\pi}\Gamma^2(\tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu}), \qquad c_- = c_+^*, \qquad |c_+| = \frac{1}{2\cosh(\pi\tilde{\nu})}. \tag{44}$$

At $|p\tau_1|, |p\tau_2| \ll 1$, the required nonlocal terms are

$$
\begin{aligned}
[D_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}} \left[\sigma_p + \frac{\mathrm{i}p(\tau_1-\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right], \\
[\widehat{D}_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} \eta c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}} \left[\sigma_p + \frac{\mathrm{i}p(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right]\boldsymbol{\epsilon}, \\
[\check{D}_{\mathsf{ab}}]_{\mathrm{NonL}} &= -\frac{1}{2}\sum_{\eta=\pm1} \eta c_\eta (p^2\tau_1\tau_2)^{\mathrm{i}\eta\tilde{\nu}}\boldsymbol{\epsilon} \left[\sigma_p - \frac{\mathrm{i}p(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde{\nu}}\mathbf{1}_2 + \cdots\right].
\end{aligned}
\tag{45}
$$

The omitted terms are second order in the soft momentum at fixed times. These nonlocal expressions are independent of the SK indices. They allow the two soft lines to be cut, while the remaining hard lines retain their full time dependence and SK ordering [4, 6].

The Yukawa interaction gives two vertex types, labelled $r = \mathrm{L}, \overline{\mathrm{L}}$ for $\psi\psi$ and $\psi^\dagger\psi^\dagger$, respectively. Their amputated rules are $-\mathsf{i}\mathsf{a}g_t a(\tau) J_r$, where $J_{\mathrm{L}} = \boldsymbol{\epsilon}$ and $J_{\overline{\mathrm{L}}} = -\boldsymbol{\epsilon}$.

3.2 The Box Diagram and Soft Expansion

We first consider the box shown in Fig. 9, with ordering 1234, for a single Majorana sector and a fixed color. We sum over all assignments of the two Yukawa vertex types at its four vertices, including the corresponding propagator combinations. External-momentum permutations will be included after this vertex-type sum.

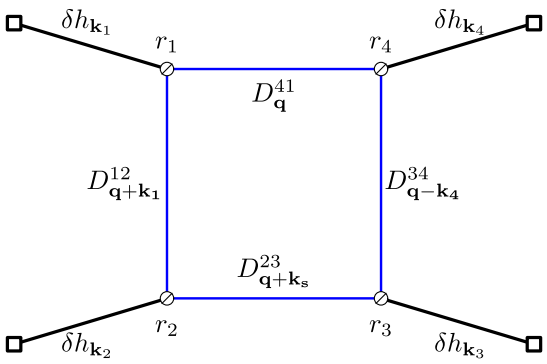


Figure 9: The top-quark box with cyclic ordering 1234. Black solid lines denote external Higgs modes and blue solid lines denote fermion propagators. The labels r_i specify the Yukawa vertex types, which are summed independently at each vertex.

For the collapsed s-channel contribution, we retain the original term together with those obtained by exchanging $\mathbf{k}_1 \leftrightarrow \mathbf{k}_2$, exchanging $\mathbf{k}_3 \leftrightarrow \mathbf{k}_4$, and performing both exchanges. The two single exchanges describe the same additional connection, while the simultaneous exchange describes the original connection traversed in reverse. Reversing the traversal does not generate new Wick pairings, so this prescription represents each set of pairings twice. We therefore multiply

the four-term sum by 1/2. With this normalization, the sum over vertex types for ordering 1234 is

$$\langle \delta h^4 \rangle'_{1234} = -\frac{g_t^4}{2} \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4=\pm 1} \left(\prod_{v=1}^{4} \mathsf{a}_v \right) \int_{-\infty}^{0} \prod_{v=1}^{4} [\mathrm{d}\tau_v \, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \\ \times \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \sum_{r_1,\ldots,r_4} \mathrm{tr}_2[D^{12} J_{r_2} D^{23} J_{r_3} D^{34} J_{r_4} D^{41} J_{r_1}], \tag{46}$$

where $D^{ij} = D^{r_i r_j}_{\mathsf{a}_i \mathsf{a}_j}(\mathbf{p}_{ij};\tau_i,\tau_j)$ and tr_2 contracts the two spinor components. The sum over $r_1,\ldots,r_4$ includes all assignments of the two Yukawa vertex types for this fixed ordering.

Choose the successive momenta as $\mathbf{q}+\mathbf{k}_1$, $\mathbf{K}=\mathbf{q}+\mathbf{k}_s$, $\mathbf{q}-\mathbf{k}_4$, and $\mathbf{q}$. In the double-soft region $q, K \sim k_s$, it is useful to introduce centered hard momenta,

$$\mathbf{k}_L = \frac{\mathbf{k}_1 - \mathbf{k}_2}{2}, \qquad \mathbf{k}_R = \frac{\mathbf{k}_3 - \mathbf{k}_4}{2}, \qquad \boldsymbol{\ell} = \mathbf{q} + \frac{\mathbf{k}_s}{2}. \tag{47}$$

The hard lines then carry $\mathbf{k}_L + \boldsymbol{\ell}$ and $\mathbf{k}_R + \boldsymbol{\ell}$. In particular the right momentum is exactly $\mathbf{q} - \mathbf{k}_4 = \mathbf{q} + \mathbf{k}_s + \mathbf{k}_3$. At leading collapsed order, $k_L \simeq k_{12}/2$ and $k_R \simeq k_{34}/2$.

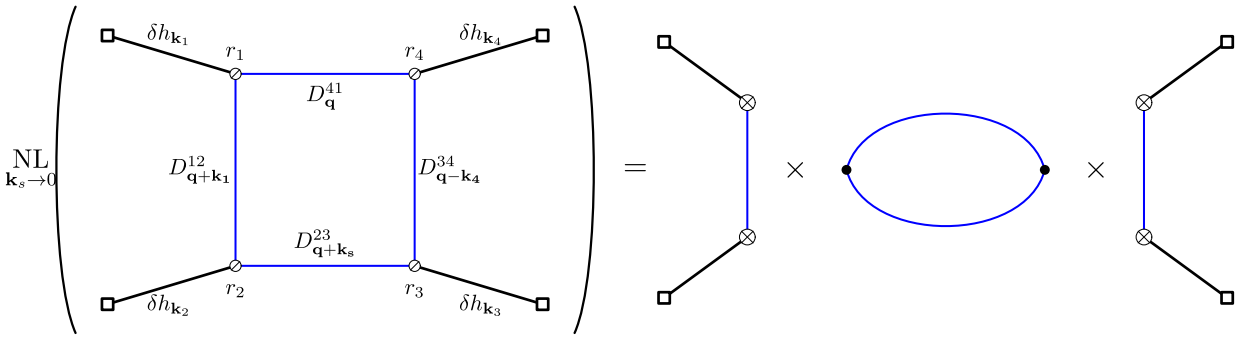


Figure 10: The leading nonlocal part of the top-quark box with cyclic ordering 1234. Cutting the two soft lines separates two tree subgraphs, each containing one hard fermion propagator. The central factor contains the nonlocal parts of the two soft propagators and their loop momentum integral. Spinor indices are contracted between the three factors.

The corresponding factorization is shown in Fig. 10. We take the branch $\eta = +1$ for example, and attaching the soft endpoints to each hard line gives matrices V_{12} and V_{34}. Their normalization is fixed in App. B.2 so that the cut spin factor of this ordering is

$$c_+^2 (qK)^{2\mathrm{i}\tilde{\nu}} \prod_{v=1}^{4} (-\tau_v)^{\mathrm{i}\tilde{\nu}} \, \mathrm{tr}_2[V_{12}\sigma_K V_{34}\sigma_q]. \tag{48}$$

The four orderings contributing to the collapsed s channel are 1234, 1243, 1342, 1432. They replace the trace by

$$\mathrm{tr}_2[(V_{12}+V_{21})\sigma_K (V_{34}+V_{43})\sigma_q]. \tag{49}$$

Every time and SK label remains attached to its physical external vertex, so all four terms have the same exact Higgs factors.

At zeroth soft order, the vertex-type sum at a hard end is proportional to its direction. Writing its scalar coefficient as $F_{0;\mathsf{ab}}$, one finds

$$\begin{aligned} V_{12}^{(0)} &= -F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, & V_{21}^{(0)} &= +F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, \\ V_{12}^{(0)} + V_{21}^{(0)} &= 0, & V_{34}^{(0)} + V_{43}^{(0)} &= 0, \end{aligned} \tag{50}$$

where $\mathbf{n}_L = \widehat{\mathbf{k}}_L$ and $\mathbf{n}_R = \widehat{\mathbf{k}}_R$. The scalar coefficient is invariant under reversal of its two time and SK labels. Thus the cancellation holds at the integrand level. A single ordering has a nonzero zeroth-order trace, but the sum removes the candidate $k_s^{3\pm 4\mathrm{i}\tilde{\nu}}$ oscillatory signal.

The first-order corrections to both the hard and soft propagators contribute to the hard-end matrices. Their sum can be organized as

$$\begin{aligned} V_{12}^{(1)} + V_{21}^{(1)} &= -\frac{2}{k_L}\boldsymbol{\sigma}\cdot\mathbf{v}_L, & \mathbf{v}_L &= \Pi_1^L\boldsymbol{\ell} + \Pi_2^L\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}), \\ V_{34}^{(1)} + V_{43}^{(1)} &= -\frac{2}{k_R}\boldsymbol{\sigma}\cdot\mathbf{v}_R, & \mathbf{v}_R &= \Pi_1^R\boldsymbol{\ell} + \Pi_2^R\mathbf{n}_R(\mathbf{n}_R\cdot\boldsymbol{\ell}). \end{aligned} \tag{51}$$

The two scalar functions Π_i depend on the hard momentum, the endpoint times and SK labels, and their explicit expressions are given in (106). The function Π_1 includes the hard-direction correction and the two soft-endpoint corrections, and Π_2 contains the hard-magnitude derivative and its associated direction correction. Substituting (51) into (49) and using $\mathrm{tr}_2[(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b})(\boldsymbol{\sigma}\cdot\mathbf{c})(\boldsymbol{\sigma}\cdot\mathbf{d})] = 2[(\mathbf{a}\cdot\mathbf{b})(\mathbf{c}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) + (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})]$, the leading term of the sum of the four traces becomes:

$$\begin{aligned} &\mathrm{tr}_2[(V_{12}^{(1)} + V_{21}^{(1)})\sigma_K(V_{34}^{(1)} + V_{43}^{(1)})\sigma_q] \\ &\quad = \frac{8}{k_L k_R q K}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\right]. \end{aligned} \tag{52}$$

The factor 8 consists of 2 from the Pauli identity and 4 from the two ordering sums. Consequently the first nonzero oscillatory signal scales as $k_s^{5\pm 4\mathrm{i}\tilde{\nu}}$.

3.3 Factorized Loop and Time Integrals

The scalar trace in (52) reduces the remaining loop to a scalar integral. We define its dimensionless coefficients by

$$\begin{aligned} &\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(qK)^{2\mathrm{i}\tilde{\nu}-1}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\right] \\ &\qquad = k_s^{5+4\mathrm{i}\tilde{\nu}}\sum_{i,j=1}^{2}\mathcal{I}_{\mathrm{top}}^{ij}(-\mathrm{i}\tilde{\nu},\theta_1,\theta_3,\phi_3)\Pi_i^L\Pi_j^R. \end{aligned} \tag{53}$$

Then we will complete this loop integral and obtain the non-analytic term of k_s. The resulting coefficients satisfy

$$\mathcal{I}_{\mathrm{top}}^{11} = 3\mathcal{I}_{\mathrm{top}}^{12} = 3\mathcal{I}_{\mathrm{top}}^{21}. \tag{54}$$

These three coefficients are independent of angles. All angular dependence is carried by $\mathcal{I}_{\mathrm{top}}^{22}$, whose explicit expression is collected in App. B.2.

For the same-branch oscillatory contribution, the nonlocal part of the loop integral in (46), summed over the four orderings, is

$$\left[\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \sum_{\mathrm{ord}} \sum_{r_1,\ldots,r_4} \mathrm{tr}_2[D^{12} J_{r_2} D^{23} J_{r_3} D^{34} J_{r_4} D^{41} J_{r_1}]\right]_{\mathrm{NonL}}$$
$$\simeq \frac{8}{k_L k_R} \sum_{\eta=\pm1} c_\eta^2 k_s^{5+4\mathrm{i}\eta\tilde{\nu}} \prod_{v=1}^{4} (-\tau_v)^{\mathrm{i}\eta\tilde{\nu}} \sum_{i,j=1}^{2} \mathcal{I}^{ij}_{\mathrm{top}}(-\mathrm{i}\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3) \Pi^L_{i,\eta} \Pi^R_{j,\eta}. \tag{55}$$

Here $\sum_{\mathrm{ord}}$ acts on the loop as in (49), and c_η is defined in (44). The kernels $\Pi_{i,\eta}$ are given explicitly in (106).

The two time integrations on each side are now independent. For the left subgraph, the leading collapsed term at $k_1 \simeq k_2 \simeq k_L$ defines

$$\sum_{\mathsf{a},\mathsf{b}=\pm1} \mathsf{ab} \int_{-\infty}^{0} \frac{\mathrm{d}\tau_1}{H\tau_1} \frac{\mathrm{d}\tau_2}{H\tau_2} (-\tau_1)^{\mathrm{i}\eta\tilde{\nu}} (-\tau_2)^{\mathrm{i}\eta\tilde{\nu}} G_{\mathsf{a}}(k_1;\tau_1) G_{\mathsf{b}}(k_2;\tau_2) \Pi_{i,\eta;\mathsf{ab}}(k_L;\tau_1,\tau_2)$$
$$\simeq \frac{H^2}{4k_1^3 k_2^3} k_L^{-2\mathrm{i}\eta\tilde{\nu}} \mathcal{P}_{\mathrm{top},i,\eta}(\tilde{\nu}), \qquad i=1,2, \quad \eta=\pm1. \tag{56}$$

Below, $\mathcal{P}_{\mathrm{top},i} \equiv \mathcal{P}_{\mathrm{top},i,+}$ denotes the positive-branch coefficient. The right subgraph follows by $(1,2,L) \to (3,4,R)$. The exact external propagators were common factors throughout the cancellation and loop integral. Corrections to their equal-energy limit enter beyond the leading nonzero term.

These time integrals reduce to the Whittaker seed of Ref. [3], denoted here by $\mathcal{K}^{\kappa;p_1p_2}_{\mathsf{ab}}$ with $\kappa = \pm 1/2$. This plays the same role for $\mathcal{P}_{\mathrm{top},i}$ as $\mathcal{M}^{p_1p_2}_{\mathsf{ab}}$ does for the gauge-boson coefficients. Its scalar kernel has a different normalization from $D^{(\mathrm{s})}$ in (6). The seed normalization, its closed form and the combinations giving $\mathcal{P}_{\mathrm{top},i}$ are given in App. B.3. A common endpoint regulator is removed after the complete SK sum has been assembled.

3.4 Oscillatory Signal

We now restore the sector and color multiplicities in the final signals. The two identical Majorana sectors supply a factor of 2, and the color sum supplies N_c, giving an overall factor $2N_c$ relative to the single-sector, fixed-color normalization in (46). Combining the soft loop and the two hard subgraphs, and adding the complex-conjugate branch, gives

$$\left[\langle \delta h^4 \rangle'_{\mathrm{top},s}\right]_{\mathrm{osc}} \simeq -\frac{N_c g_t^4 H^4 k_s^5}{16 k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R}$$
$$\times \left[\left(\frac{k_s^2}{4k_{12}k_{34}}\right)^{2\mathrm{i}\tilde{\nu}} C_{\mathrm{top}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \mathrm{c.c.}\right], \tag{57}$$

where

$$C_{\mathrm{top}} = 8c_+^2\, 16^{2\mathrm{i}\tilde{\nu}} \sum_{i,j=1}^{2} \mathcal{P}_{\mathrm{top},i} \mathcal{P}_{\mathrm{top},j} \mathcal{I}^{ij}_{\mathrm{top}}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3). \tag{58}$$

All four s-channel orderings are included. The phase factor $16^{2\mathrm{i}\tilde{\nu}}$ converts the natural hard scale $k_L k_R$ to the momentum ratio used in Sec. 2, since $4k_{12}k_{34} \simeq 16 k_L k_R$.

The angular decomposition takes the same form as (37):

$$C_{\rm top} = A_{1,\rm top} + A_{2,\rm top}(\cos^2\theta_1 + \cos^2\theta_3) + A_{3,\rm top}\cos^2\theta_1\cos^2\theta_3 + B_{\rm top}\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 + D_{\rm top}\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3). \tag{59}$$

The five coefficients depend only on $\tilde{\nu}$ and satisfy $A_{3,\rm top} = -3A_{2,\rm top}$. Their analytic expressions are given in App. B.4. In particular all nonconstant angular terms are proportional to $\mathcal{P}^2_{\rm top,2}$. The result is even under either Higgs-pair exchange and under parity.

At fixed angles the oscillatory bracket in (57) is

$$2|C_{\rm top}|\cos\left[2\tilde{\nu}\log\left(\frac{k_s^2}{4k_{12}k_{34}}\right) + \arg C_{\rm top}\right]. \tag{60}$$

The frequency in $\log k_s$ is $4m_t/H$. Compared with the gauge-boson signals, the top oscillation carries an additional soft suppression $k_s^2/(k_L k_R)$ from (50). The coefficient determines its mass-dependent and angle-dependent amplitude and phase.

4 Conclusions

A Gauge Boson Integrals and Coefficients

A.1 Loop Integral Formulae

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ reduces all angular moments in (15) to the scalar integral f_0:

$$f_r(\alpha,\beta) = \frac{1}{2^r}\sum_{u+v+w=r}\frac{r!}{u!v!w!}(-1)^{v+w} f_0\left(\alpha + \frac{r}{2} - v, \beta - u\right), \tag{61}$$

where the sum runs over nonnegative integers. For the third and fourth moments this gives

$$\begin{aligned} f_3(\alpha,\beta) = \frac{1}{8}\Big[&f_0(\alpha+\tfrac{3}{2},\beta-3) - 3f_0(\alpha+\tfrac{3}{2},\beta-2) - 3f_0(\alpha+\tfrac{1}{2},\beta-2) \\ &+ 3f_0(\alpha+\tfrac{3}{2},\beta-1) + 6f_0(\alpha+\tfrac{1}{2},\beta-1) + 3f_0(\alpha-\tfrac{1}{2},\beta-1) \\ &- f_0(\alpha+\tfrac{3}{2},\beta) - 3f_0(\alpha+\tfrac{1}{2},\beta) - 3f_0(\alpha-\tfrac{1}{2},\beta) - f_0(\alpha-\tfrac{3}{2},\beta)\Big], \end{aligned} \tag{62}$$

$$\begin{aligned} f_4(\alpha,\beta) = \frac{1}{16}\Big[&f_0(\alpha+2,\beta-4) - 4f_0(\alpha+2,\beta-3) - 4f_0(\alpha+1,\beta-3) \\ &+ 6f_0(\alpha+2,\beta-2) + 12f_0(\alpha+1,\beta-2) + 6f_0(\alpha,\beta-2) \\ &- 4f_0(\alpha+2,\beta-1) - 12f_0(\alpha+1,\beta-1) - 12f_0(\alpha,\beta-1) \\ &- 4f_0(\alpha-1,\beta-1) + f_0(\alpha+2,\beta) + 4f_0(\alpha+1,\beta) \\ &+ 6f_0(\alpha,\beta) + 4f_0(\alpha-1,\beta) + f_0(\alpha-2,\beta)\Big]. \end{aligned} \tag{63}$$

A.2 Bubble Coefficients

For $x = -i\eta\tilde{\nu}$, the relative weight of a longitudinal nonlocal propagator is

$$\beta(x) \equiv \frac{(x+1/2)^2}{1/4 - x^2} = \frac{1+2x}{1-2x}. \tag{64}$$

Define $\mathcal{I}_{\sigma_1\sigma_2}(x)$ by restricting the loop in (22) to the polarizations (σ_1, σ_2) and replacing $\mathcal{I}_{\rm Bub}$ on its right-hand side by $\mathcal{I}_{\sigma_1\sigma_2}$. The independent coefficients are

$$\mathcal{I}_{LL}(x) = [\beta(x)]^2 \left[\mathcal{J}_0(2,2;x) + 2\mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x)\right], \tag{65}$$

$$\mathcal{I}_{L+}(x) = \frac{\beta(x)}{2}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right], \tag{66}$$

$$\begin{aligned}\mathcal{I}_{++}(x) = \frac{1}{4}\big[&\mathcal{J}_0(0,0;x) + \mathcal{J}_0(2,2;x) + \mathcal{J}_2(0,2;x) \\ &+2\mathcal{J}_0(1,1;x) + 2\mathcal{J}_1(1,2;x) + 2\mathcal{J}_1(0,1;x)\big],\end{aligned} \tag{67}$$

$$\begin{aligned}\mathcal{I}_{+-}(x) = \frac{1}{4}\big[&\mathcal{J}_0(0,0;x) + \mathcal{J}_0(2,2;x) + \mathcal{J}_2(0,2;x) \\ &-2\mathcal{J}_0(1,1;x) + 2\mathcal{J}_1(1,2;x) - 2\mathcal{J}_1(0,1;x)\big].\end{aligned} \tag{68}$$

The remaining entries satisfy $\mathcal{I}_{--} = \mathcal{I}_{++}$, $\mathcal{I}_{-+} = \mathcal{I}_{+-}$, and $\mathcal{I}_{L-} = \mathcal{I}_{+L} = \mathcal{I}_{-L} = \mathcal{I}_{L+}$. Consequently,

$$\mathcal{I}_{\rm Bub} = \mathcal{I}_{LL} + 4\mathcal{I}_{L+} + 2\mathcal{I}_{++} + 2\mathcal{I}_{+-}, \tag{69}$$

which reduces to (23) using the moment relations in App. A.1.

For the $\eta = +1$ branch, the fully reduced sum is

$$\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}) = \frac{3\cdot 2^{-4-4\mathrm{i}\tilde{\nu}}(3\mathrm{i}-2\tilde{\nu})^2\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(3+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi[1+(3\mathrm{i}-2\tilde{\nu})\tilde{\nu}]^2}. \tag{70}$$

Substituting this into (27) gives

$$\begin{aligned}C^{\rm Bub}(\tilde{\nu}) = &\frac{3(\tilde{\nu}-\mathrm{i})(\tilde{\nu}-2\mathrm{i})^4}{256\pi^{7/2}\tilde{\nu}^2(2\tilde{\nu}-\mathrm{i})}\Gamma^2(-2-\mathrm{i}\tilde{\nu})\Gamma^2(5/2+\mathrm{i}\tilde{\nu}) \\ &\times\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(1+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu}).\end{aligned} \tag{71}$$

For real $\tilde{\nu} > 0$, the opposite nonlocal branch is their complex conjugate.

A.3 Triangle Coefficients

A.3.1 Polarization sum

Let $x = -\mathrm{i}\eta\tilde{\nu}$ and let $\beta(x)$ be the longitudinal weight in (64). Summing the nine soft-polarization combinations gives

$$\begin{aligned}\mathcal{I}^L_{\rm Tri}(x,\theta) = [\beta(x)]^2 \Big\{&\frac{\sin^2\theta}{2}\left[\mathcal{J}_0(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_2(2,2;x) - \mathcal{J}_3(1,2;x)\right] \\ &+\cos^2\theta\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x)\right]\Big\} \\ &+\frac{\beta(x)(1+\cos^2\theta)}{2}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right] \\ &+\frac{\sin^2\theta}{2}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_2(0,2;x) + \mathcal{J}_0(0,0;x)\right] \\ &+\cos^2\theta\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right].\end{aligned} \tag{72}$$

The terms proportional to β^2, β, and 1 arise from two longitudinal, mixed, and two transverse soft lines, respectively.

For a hard line with helicity +, the corresponding expression is

$$\begin{aligned}\mathcal{I}^{+}_{\rm Tri}(x,\theta) = [\beta(x)]^2 \Bigg\{ &\frac{1+\cos^2\theta}{4}\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right] \\ &+\frac{\sin^2\theta}{2}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_2(0,2;x)\right]\Bigg\} \\ &+ \frac{\beta(x)(2+\sin^2\theta)}{4}\left[\mathcal{J}_0(0,2;x) - \mathcal{J}_2(0,2;x)\right] \\ &+ \frac{1+\cos^2\theta}{4}\left[\mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) + \mathcal{J}_3(1,2;x) + \mathcal{J}_2(0,2;x) + \mathcal{J}_0(0,0;x)\right] \\ &+ \frac{\sin^2\theta}{2}\left[\mathcal{J}_0(2,2;x) - \mathcal{J}_2(2,2;x) + \mathcal{J}_1(1,2;x) - \mathcal{J}_3(1,2;x)\right]. \end{aligned} \tag{73}$$

At zero chemical potential, reducing the Gamma functions gives the same result for all three hard-line polarizations:

$$\begin{aligned}\mathcal{I}^{L}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) = \mathcal{I}^{+}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) = \mathcal{I}^{-}_{\rm Tri}(-\mathrm{i}\tilde{\nu},\theta) &= \frac{1}{3}\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}) \\ &= \frac{2^{-4-4\mathrm{i}\tilde{\nu}}(3\mathrm{i}-2\tilde{\nu})^2\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(3+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi[1+(3\mathrm{i}-2\tilde{\nu})\tilde{\nu}]^2}. \end{aligned} \tag{74}$$

Summing over the hard-line polarizations reproduces $\mathcal{I}_{\rm Bub}$ by polarization completeness. Thus the angular dependence of the separate soft-polarization contributions cancels in their sum. The coefficient of a single triangle reduces to

$$C^{\rm Tri}(\tilde{\nu}) = \frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{3(4\pi)^{7/2}}\mathcal{P}^{(1)}_{+}(\tilde{\nu})\left[\mathcal{P}^{(2),L}_{+}(\tilde{\nu}) + 2\mathcal{P}^{(2),+}_{+}(\tilde{\nu})\right]\mathcal{I}_{\rm Bub}(-\mathrm{i}\tilde{\nu}). \tag{75}$$

A.3.2 Time-integral coefficients

For the scalar propagator of (6), the double-folded seed is defined by

$$\mathcal{M}^{p_1p_2}_{\mathsf{ab}} \equiv (-\mathsf{ab})k^{5+p_{12}}\int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{p_1}(-\tau_2)^{p_2}e^{\mathrm{i}\mathsf{a}k\tau_1+\mathrm{i}\mathsf{b}k\tau_2}\frac{D^{(\mathrm{s})}_{\mathsf{ab}}(k;\tau_1,\tau_2)}{H^2}, \tag{76}$$

This is the seed $\widetilde{\mathcal{I}}^{p_1p_2}_{\mathsf{ab}}(1,1)$ of Ref. [3], with H restored explicitly. All seeds below have mass index $\tilde{\nu}$. The closed expressions are

$$\mathcal{M}^{p_1p_2}_{\pm\mp} = \frac{e^{\mp\mathrm{i}\pi\bar{p}_{12}/2}}{2^{5+p_{12}}}\prod_{j=1}^{2}\frac{\Gamma(5/2+p_j-\mathrm{i}\tilde{\nu})\Gamma(5/2+p_j+\mathrm{i}\tilde{\nu})}{\Gamma(3+p_j)}, \tag{77}$$

$$\begin{aligned}\mathcal{M}^{p_1p_2}_{\pm\pm} = &\frac{\pm\mathrm{i}e^{\mp\mathrm{i}\pi p_{12}/2}e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}}\prod_{j=1}^{2}\frac{\Gamma(5/2+p_j-\mathrm{i}\tilde{\nu})\Gamma(5/2+p_j+\mathrm{i}\tilde{\nu})}{\Gamma(3+p_j)} \\ &-\frac{e^{\mp\mathrm{i}\pi p_{12}/2}}{2^{5+p_{12}}}\Gamma(5+p_{12})\Gamma(5/2+p_1\pm\mathrm{i}\tilde{\nu})\Gamma(5/2+p_2\pm\mathrm{i}\tilde{\nu}) \\ &\times {}_3\widetilde{F}_2\left[\begin{matrix}5+p_{12},\ 1/2\pm\mathrm{i}\tilde{\nu},\ 1\\ 7/2+p_1\pm\mathrm{i}\tilde{\nu},\ 7/2+p_2\pm\mathrm{i}\tilde{\nu}\end{matrix}\middle|1\right]. \end{aligned} \tag{78}$$

Here ${}_3\widetilde{F}_2(a,b,c;d,e;1) = {}_3F_2(a,b,c;d,e;1)/[\Gamma(d)\Gamma(e)]$ is the regularized hypergeometric function.

Set $p_\epsilon = -5/2 + \epsilon + \mathrm{i}\eta\tilde{\nu}$. The two transverse coefficients are

$$\mathcal{P}_\eta^{(2),\pm}(\tilde{\nu}) = 2^{2\mathrm{i}\eta\tilde{\nu}} \mathrm{FP}_{\epsilon=0} \sum_{\mathsf{a},\mathsf{b}=\pm 1} \left[\mathcal{M}_{\mathsf{ab}}^{p_\epsilon,p_\epsilon} + \mathsf{ia}\mathcal{M}_{\mathsf{ab}}^{p_\epsilon+1,p_\epsilon} + \mathsf{ib}\mathcal{M}_{\mathsf{ab}}^{p_\epsilon,p_\epsilon+1} - \mathsf{ab}\mathcal{M}_{\mathsf{ab}}^{p_\epsilon+1,p_\epsilon+1} \right]. \tag{79}$$

The longitudinal coefficient can be written using the same transverse combination:

$$\begin{aligned}\mathcal{P}_\eta^{(2),L}(\tilde{\nu}) = {} & \frac{\mathcal{P}_\eta^{(1)}(\tilde{\nu}) + (1/2 + \mathrm{i}\eta\tilde{\nu})^2 \mathcal{P}_\eta^{(2),+}(\tilde{\nu})}{\tilde{\nu}^2 + 1/4} \\ & + \frac{2^{2\mathrm{i}\eta\tilde{\nu}}}{\tilde{\nu}^2 + 1/4} \mathrm{FP}_{\epsilon=0} \sum_{\mathsf{a},\mathsf{b}=\pm 1} \Big\{ (1/2 + \mathrm{i}\eta\tilde{\nu}) \left[\mathcal{M}_{\mathsf{ab}}^{p_\epsilon+2,p_\epsilon} + \mathsf{ib}\mathcal{M}_{\mathsf{ab}}^{p_\epsilon+2,p_\epsilon+1} \right. \\ & \left. + \mathcal{M}_{\mathsf{ab}}^{p_\epsilon,p_\epsilon+2} + \mathsf{ia}\mathcal{M}_{\mathsf{ab}}^{p_\epsilon+1,p_\epsilon+2} \right] + \mathcal{M}_{\mathsf{ab}}^{p_\epsilon+2,p_\epsilon+2} \Big\}. \end{aligned} \tag{80}$$

The operator $\mathrm{FP}_{\epsilon=0}$ selects the constant coefficient of the Laurent expansion after assembling the indicated seed sum. The first term proportional to $\mathcal{P}_\eta^{(1)}$ in (80) is the contact contribution from differentiating the time-ordering step functions.

A.4 Box Coefficients

A.4.1 Polarization sums

Let $\lambda_1, \lambda_3 = L, \pm$ denote the polarizations of the hard lines with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$. With $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, define the soft-polarization sum by

$$\begin{aligned}& \sum_{\sigma_1,\sigma_2=L,\pm} [\beta(x)]^{\delta_{\sigma_1 L} + \delta_{\sigma_2 L}} \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \left(\frac{q}{2}\right)^{-2x} \left(\frac{K}{2}\right)^{-2x} (e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)})(e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_3)}) \\ & \quad \times (e_{\mathbf{k}_3}^{(\lambda_3)*} \cdot e_{\mathbf{q}}^{(\sigma_2)})(e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}) = \frac{4^{2x} k_s^{3-4x}}{(4\pi)^{3/2}} \mathcal{I}_{\mathrm{Box}}^{\lambda_1\lambda_3}(x, \theta_1, \theta_3, \phi_3). \end{aligned} \tag{81}$$

The angles are defined by the convention stated below (1). The two transverse hard-line coefficients $\mathcal{P}_\eta^{(2),+} = \mathcal{P}_\eta^{(2),-}$ are equal, so it suffices to retain three groups:

$$\mathcal{I}_{\mathrm{Box}}^{LH} \equiv \sum_{\lambda=\pm} \left(\mathcal{I}_{\mathrm{Box}}^{L\lambda} + \mathcal{I}_{\mathrm{Box}}^{\lambda L} \right), \qquad \mathcal{I}_{\mathrm{Box}}^{HH} \equiv \sum_{\lambda_1,\lambda_3=\pm} \mathcal{I}_{\mathrm{Box}}^{\lambda_1\lambda_3}, \tag{82}$$

together with $\mathcal{I}_{\mathrm{Box}}^{LL}$. Here H in a polarization superscript denotes the sum over transverse helicities.

For $x = -\mathrm{i}\tilde{\nu}$, the longitudinal result is

$$\begin{aligned}& \mathcal{I}_{\mathrm{Box}}^{LL}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\ & \quad = -\frac{4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu} - 3\mathrm{i})\Gamma(-3/2 - 2\mathrm{i}\tilde{\nu})\Gamma(1 + 2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - \mathrm{i})(\tilde{\nu} - 2\mathrm{i})(2\tilde{\nu} - \mathrm{i})} \\ & \qquad \times \Big\{ -16 - 19\mathrm{i}\tilde{\nu} + 6\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - \mathrm{i}) \Big[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3) \Big] \\ & \qquad\quad + 4(2\tilde{\nu}^2 - 5\mathrm{i}\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1 \sin^2\theta_3 \\ & \qquad\quad - 8\tilde{\nu}(\tilde{\nu} - \mathrm{i})\cos\phi_3 \sin(2\theta_1)\sin(2\theta_3) \Big\}. \end{aligned} \tag{83}$$

Reducing the mixed and transverse results to the same Gamma-function basis gives

$$\begin{aligned}
&\mathcal{I}^{LH}_{\rm Box}(-\mathrm{i}\tilde{\nu},\theta_1,\theta_3,\phi_3)\\
&\quad=-\frac{2\cdot 4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu}-3\mathrm{i})\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(1+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu}-\mathrm{i})(\tilde{\nu}-2\mathrm{i})(2\tilde{\nu}-\mathrm{i})}\\
&\qquad\times\Big\{-32-37\mathrm{i}\tilde{\nu}+10\tilde{\nu}^2-\tilde{\nu}(2\tilde{\nu}-\mathrm{i})\Big[\cos(2\theta_1)+\cos(2\theta_3)+3\cos(2\theta_1)\cos(2\theta_3)\Big]\\
&\qquad\qquad-4(2\tilde{\nu}^2-5\mathrm{i}\tilde{\nu}-3)\cos(2\phi_3)\sin^2\theta_1\sin^2\theta_3\\
&\qquad\qquad+8\tilde{\nu}(\tilde{\nu}-\mathrm{i})\cos\phi_3\sin(2\theta_1)\sin(2\theta_3)\Big\},
\end{aligned}\tag{84}$$

$$\begin{aligned}
&\mathcal{I}^{HH}_{\rm Box}(-\mathrm{i}\tilde{\nu},\theta_1,\theta_3,\phi_3)\\
&\quad=-\frac{4^{-3-2\mathrm{i}\tilde{\nu}}(2\tilde{\nu}-3\mathrm{i})\Gamma(-3/2-2\mathrm{i}\tilde{\nu})\Gamma(1+2\mathrm{i}\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu}-\mathrm{i})(\tilde{\nu}-2\mathrm{i})(2\tilde{\nu}-\mathrm{i})}\\
&\qquad\times\Big\{-64-75\mathrm{i}\tilde{\nu}+22\tilde{\nu}^2+\tilde{\nu}(2\tilde{\nu}-\mathrm{i})\Big[\cos(2\theta_1)+\cos(2\theta_3)+3\cos(2\theta_1)\cos(2\theta_3)\Big]\\
&\qquad\qquad+4(2\tilde{\nu}^2-5\mathrm{i}\tilde{\nu}-3)\cos(2\phi_3)\sin^2\theta_1\sin^2\theta_3\\
&\qquad\qquad-8\tilde{\nu}(\tilde{\nu}-\mathrm{i})\cos\phi_3\sin(2\theta_1)\sin(2\theta_3)\Big\}.
\end{aligned}\tag{85}$$

These expressions satisfy the relations implied by polarization completeness and (74):

$$\mathcal{I}^{LH}_{\rm Box}=\frac{2}{3}\mathcal{I}_{\rm Bub}-2\mathcal{I}^{LL}_{\rm Box},\tag{86}$$

$$\mathcal{I}^{HH}_{\rm Box}=\frac{1}{3}\mathcal{I}_{\rm Bub}+\mathcal{I}^{LL}_{\rm Box}.\tag{87}$$

Indeed, all angular terms cancel in $\mathcal{I}^{LH}_{\rm Box}+2\mathcal{I}^{LL}_{\rm Box}$ and $\mathcal{I}^{HH}_{\rm Box}-\mathcal{I}^{LL}_{\rm Box}$. The three groups therefore also satisfy $\mathcal{I}^{LL}_{\rm Box}+\mathcal{I}^{LH}_{\rm Box}+\mathcal{I}^{HH}_{\rm Box}=\mathcal{I}_{\rm Bub}$.

The dimensionless box coefficient is consequently

$$\begin{aligned}
C^{\rm Box}(\tilde{\nu},\theta_1,\theta_3,\phi_3)&=\frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\Big\{[\mathcal{P}^{(2),L}_{+}]^2\mathcal{I}^{LL}_{\rm Box}+\mathcal{P}^{(2),L}_{+}\mathcal{P}^{(2),+}_{+}\mathcal{I}^{LH}_{\rm Box}+[\mathcal{P}^{(2),+}_{+}]^2\mathcal{I}^{HH}_{\rm Box}\Big\}\\
&=\frac{\Gamma^4(-\mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}}\Big\{[\mathcal{P}^{(2),L}_{+}-\mathcal{P}^{(2),+}_{+}]^2\mathcal{I}^{LL}_{\rm Box}+\frac{1}{3}\mathcal{P}^{(2),+}_{+}[2\mathcal{P}^{(2),L}_{+}+\mathcal{P}^{(2),+}_{+}]\mathcal{I}_{\rm Bub}\Big\},
\end{aligned}\tag{88}$$

The opposite nonlocal branch follows by complex conjugation.

A.4.2 Angular projections

The five functions in (37) can be read off from (88), or equivalently obtained from the following projections:

$$A_{1,g}(\tilde{\nu}) = \frac{1}{2}\left[C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{2}, \tfrac{\pi}{2}, 0) + C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{2}, \tfrac{\pi}{2}, \tfrac{\pi}{2})\right], \tag{89}$$

$$A_{2,g}(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{2}, 0, 0) - A_{1,g}(\tilde{\nu}), \tag{90}$$

$$A_{3,g}(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, 0, 0, 0) - A_{1,g}(\tilde{\nu}) - 2A_{2,g}(\tilde{\nu}), \tag{91}$$

$$B_g(\tilde{\nu}) = \frac{1}{2}\left[C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{4}, \tfrac{\pi}{4}, 0) - C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{4}, \tfrac{\pi}{4}, \pi)\right], \tag{92}$$

$$D_g(\tilde{\nu}) = \frac{1}{2}\left[C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{2}, \tfrac{\pi}{2}, 0) - C^{\text{Box}}(\tilde{\nu}, \tfrac{\pi}{2}, \tfrac{\pi}{2}, \tfrac{\pi}{2})\right]. \tag{93}$$

B Top Quark Integrals and Coefficients

B.1 Propagators and Spinor Contractions

We use $\sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma})$, $\bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma})$, and $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta$ with $\epsilon_{12} = 1$, $\epsilon^{12} = -1$. The helicity projectors are $P_s = h^s h^{s\dagger} = (\mathbf{1}_2 + s\sigma_k)/2$. For the rescaled field, the mode expansion is

$$\psi(\tau, \mathbf{x}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \sum_{s=\pm 1} \left[h^s u_s b^s_{\mathbf{k}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} + \boldsymbol{\epsilon} h^{s*} v^*_s b^{s\dagger}_{\mathbf{k}} e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\right], \tag{94}$$

where $\{b^s_{\mathbf{k}}, b^{s'\dagger}_{\mathbf{k}'}\} = (2\pi)^3\delta^{ss'}\delta^3(\mathbf{k}-\mathbf{k}')$. The modes in (41) satisfy $\mathrm{i}\partial_\tau u_s + sku_s = am_t v_s$ and $\mathrm{i}\partial_\tau v_s - skv_s = am_t u_s$. With $u_{sj} = u_s(k, \tau_j)$ and $v_{sj} = v_s(k, \tau_j)$, the three kinds of propagators are

$$D_{-+} = \sum_s u_{s1}u^*_{s2}P_s, \qquad \widehat{D}_{-+} = -\sum_s u_{s1}v^*_{s2}P_s\boldsymbol{\epsilon}, \qquad \check{D}_{-+} = \sum_s v_{s1}u^*_{s2}\boldsymbol{\epsilon}P_s. \tag{95}$$

The signs follow from $h^s(\boldsymbol{\epsilon}h^{s*})^T = -P_s\boldsymbol{\epsilon}$.

For the seed reduction it is useful to express these propagators through the scalar propagator with chemical potential [3]. With $X = -k\tau_1$, $Y = -k\tau_2$,

$$\begin{aligned} D^{(\kappa)}_{-+}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X) W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}Y), \\ D^{(\kappa)}_{+-}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}X) W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}Y). \end{aligned} \tag{96}$$

Here $W_{-\kappa,\mathrm{i}\tilde{\nu}}$ is the Whittaker W function. The other two SK components follow from (43). In the notation of the reference, $\kappa = \mathrm{i}h\tilde{\mu}$, and in the spinor case this parameter is analytically continued to $\kappa = \pm 1/2$. We denote the kernels at these two values explicitly by $D^{(+1/2)}$ and $D^{(-1/2)}$, respectively.

Then thees three kinds of spinor propagator can be expressed as:

$$\begin{aligned} D_{\mathsf{ab}} &= \frac{\mathrm{i}k}{\sqrt{XY}}\left[(-X\partial_X - \mathrm{i}X + \tfrac{1}{2})D^{(-1/2)}_{\mathsf{ab}}P_- - (X\partial_X - \mathrm{i}X - \tfrac{1}{2})D^{(+1/2)}_{\mathsf{ab}}P_+\right], \\ \widehat{D}_{\mathsf{ab}} &= -\frac{\tilde{\nu}k}{\sqrt{XY}}[D^{(-1/2)}_{\mathsf{ab}}P_+ + D^{(+1/2)}_{\mathsf{ab}}P_-]\boldsymbol{\epsilon}, \\ \check{D}_{\mathsf{ab}} &= \frac{\tilde{\nu}k}{\sqrt{XY}}\boldsymbol{\epsilon}[D^{(+1/2)}_{\mathsf{ab}}P_+ + D^{(-1/2)}_{\mathsf{ab}}P_-]. \end{aligned} \tag{97}$$

And they satisfy:

$$D^{\mathrm{L}\overline{\mathrm{L}}} = D, \quad D^{\mathrm{LL}} = \widehat{D}, \quad D^{\overline{\mathrm{LL}}} = \check{D}, \quad D^{\overline{\mathrm{LL}}}_{\mathsf{ab}}(\mathbf{k};\tau_1,\tau_2) = -[D_{\mathsf{ba}}(-\mathbf{k};\tau_2,\tau_1)]^T. \tag{98}$$

Expanding the Whittaker W function:

$$W_{\kappa,\mathrm{i}\tilde{\nu}}(z) = \sum_{\eta=\pm1} \frac{\Gamma(-2\mathrm{i}\eta\tilde{\nu})}{\Gamma(\frac{1}{2}-\mathrm{i}\eta\tilde{\nu}-\kappa)} z^{1/2+\mathrm{i}\eta\tilde{\nu}} \left[1 - \frac{\kappa z}{1+2\mathrm{i}\eta\tilde{\nu}} + O(z^2)\right] \tag{99}$$

B.2 Soft Expansion and Loop Integrals

For the $\eta = +1$ branch, soft line can be factorized as:

$$D^{r_1 r_2}_{\mathrm{NonL},+}(p;\tau,\tau') \simeq -\frac{c_+}{2}(p^2\tau\tau')^{\mathrm{i}\tilde{\nu}} L_{r_1}(p,\tau)\sigma_p U_{r_2}(p,\tau'), \tag{100}$$

where, through first order,

$$\begin{aligned} L_{\mathrm{L}} &= \mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}, & L_{\overline{\mathrm{L}}} &= \boldsymbol{\epsilon}\left(\mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}\right), \\ U_{\mathrm{L}} &= \left(\mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}\right)\boldsymbol{\epsilon}, & U_{\overline{\mathrm{L}}} &= \mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{p}}{1+2\mathrm{i}\tilde{\nu}}. \end{aligned} \tag{101}$$

The matrices in (48) are the vertex-type sums

$$\begin{aligned} V_{12} &= -\frac{1}{2}\sum_{r_1,r_2} U_{r_1}(q,\tau_1) J_{r_1} D^{r_1 r_2}_{\mathsf{a}_1\mathsf{a}_2}(\mathbf{k}_L+\boldsymbol{\ell};\tau_1,\tau_2) J_{r_2} L_{r_2}(K,\tau_2), \\ V_{21} &= -\frac{1}{2}\sum_{r_1,r_2} U_{r_2}(q,\tau_2) J_{r_2} D^{r_2 r_1}_{\mathsf{a}_2\mathsf{a}_1}(-\mathbf{k}_L+\boldsymbol{\ell};\tau_2,\tau_1) J_{r_1} L_{r_1}(K,\tau_1). \end{aligned} \tag{102}$$

The right matrices follow by $(L,1,2,q,K) \to (R,3,4,K,q)$. The factors $-1/2$ in these definitions reproduce the normalization of the two cut lines.

At zeroth order, the vertex-type sum gives $V_{12}^{(0)} = -F_0 \boldsymbol{\sigma}\cdot\mathbf{n}_L$, with F_0 defined below. Reversal changes the direction but leaves F_0 unchanged, proving (50). At first order, the hard-line correction in V_{12} is

$$-\frac{1}{k_L}\boldsymbol{\sigma}\cdot\left[F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\ell} + (p\partial_p - 1)F_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell})\right], \tag{103}$$

while the two soft endpoints supply

$$\frac{\mathrm{i}}{1+2\mathrm{i}\tilde{\nu}}\boldsymbol{\sigma}\cdot[\tau_1 F_{1;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\mathbf{q} + \tau_2 F_{1;\mathsf{a}_2\mathsf{a}_1}(k_L;\tau_2,\tau_1)\mathbf{K}]. \tag{104}$$

The scalar coefficients follow from (97). With $X = -p\tau_1$, $Y = -p\tau_2$, their expressions for soft branch $\eta = \pm 1$ are

$$\begin{aligned} F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= \frac{\mathrm{i}p}{2\sqrt{XY}}\left[(X\partial_X - \mathrm{i}X - \tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu})D^{(+1/2)}_{\mathsf{ab}} + (-X\partial_X - \mathrm{i}X + \tfrac{1}{2} + \mathrm{i}\eta\tilde{\nu})D^{(-1/2)}_{\mathsf{ab}}\right], \\ F_{1,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= \frac{\mathrm{i}p}{2\sqrt{XY}}\left[(-X\partial_X - \mathrm{i}X + \tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu})D^{(-1/2)}_{\mathsf{ab}} - (X\partial_X - \mathrm{i}X - \tfrac{1}{2} + \mathrm{i}\eta\tilde{\nu})D^{(+1/2)}_{\mathsf{ab}}\right]. \end{aligned} \tag{105}$$

We write $F_j \equiv F_{j,+}$ for the positive branch, with $F_{0;\mathsf{ab}}(p;\tau_1,\tau_2) = F_{0;\mathsf{ba}}(p;\tau_2,\tau_1)$. The reversed ordering has the same hard correction and interchanges the coefficients of $\mathbf{q}$ and $\mathbf{K}$. Since $\mathbf{q}+\mathbf{K} = 2\boldsymbol{\ell}$, adding the two orderings gives (51). Repeating the endpoint contraction for either branch $\eta = \pm 1$ gives

$$\begin{aligned}
\Pi_{1,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) &= F_{0,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) - \frac{\mathrm{i}k}{1+2\mathrm{i}\eta\tilde{\nu}}[\tau_1 F_{1,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) + \tau_2 F_{1,\eta;\mathsf{ba}}(k;\tau_2,\tau_1)],\\
\Pi_{2,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) &= (p\partial_p - 1)F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2)|_{p=k}\,.
\end{aligned} \tag{106}$$

Setting $\eta = +1$ or -1 gives $\Pi_{i,+;\mathsf{ab}}$ or $\Pi_{i,-;\mathsf{ab}}$ at the same SK labels and endpoint times; $\Pi_{i;\mathsf{ab}} \equiv \Pi_{i,+;\mathsf{ab}}$ in the positive-branch derivation above and in the time integrals below. The momentum derivative is at fixed times. This addition involves two physical diagrams and does not average their time or SK labels.

For the loop coefficients, set $x = -\mathrm{i}\tilde{\nu}$ and $c_i = \cos\theta_i$. We use $\mathcal{J}_r(m,n;x)$ as defined in (19). The common argument x of $\mathcal{I}^{ij}_{\mathrm{top}}$ and $\mathcal{J}_r$, and the angular arguments of $\mathcal{I}^{ij}_{\mathrm{top}}$, are suppressed below. Azimuthal averaging and the scalar-convolution identities give

$$\begin{aligned}
\mathcal{I}^{11}_{\mathrm{top}} &= 3\mathcal{I}^{12}_{\mathrm{top}} = 3\mathcal{I}^{21}_{\mathrm{top}},\\
(4\pi)^{3/2}\mathcal{I}^{12}_{\mathrm{top}} &= \frac{1}{2}[\mathcal{J}_0(3,1) - \mathcal{J}_2(3,1) + \mathcal{J}_1(2,1) - \mathcal{J}_3(2,1)]\\
&\quad + \frac{1}{4}[\mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)],\\
(4\pi)^{3/2}\mathcal{I}^{22}_{\mathrm{top}} &= \frac{1}{4}[2\mathcal{J}_4(3,1) - 3\mathcal{J}_2(3,1) + \mathcal{J}_0(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)]\\
&\quad + \frac{1}{4}\Big[-6\mathcal{J}_4(3,1) + 7\mathcal{J}_2(3,1) - \mathcal{J}_0(3,1) + 5\mathcal{J}_1(2,1) - 5\mathcal{J}_3(2,1)\\
&\qquad + \mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)\Big](c_1^2 + c_3^2 - 3c_1^2c_3^2)\\
&\quad + \frac{1}{16}\Big[-16\mathcal{J}_4(3,1) + 14\mathcal{J}_2(3,1) - 2\mathcal{J}_0(3,1) + 10\mathcal{J}_1(2,1) - 18\mathcal{J}_3(2,1)\\
&\qquad + 2\mathcal{J}_0(1,1) - 7\mathcal{J}_2(1,1) - \mathcal{J}_1(0,1)\Big]\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad + \frac{1}{4}[\mathcal{J}_4(3,1) - \mathcal{J}_2(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)]\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3).
\end{aligned} \tag{107}$$

The finite sum (61) and the Gamma recurrence give the closed form below, with the common factor

$$\mathcal{G}_{\mathrm{top}}(\tilde{\nu}) = \frac{\Gamma(-\frac{1}{2} - 2\mathrm{i}\tilde{\nu})\Gamma^2(1+\mathrm{i}\tilde{\nu})}{(4\pi)^{3/2}\Gamma(2+2\mathrm{i}\tilde{\nu})\Gamma^2(\frac{1}{2}-\mathrm{i}\tilde{\nu})}. \tag{108}$$

For $x = -\mathrm{i}\tilde{\nu}$, the fully reduced coefficients are

$$\begin{aligned}
\mathcal{I}^{12}_{\rm top}(-\mathrm{i}\tilde{\nu}) &= -\mathcal{G}_{\rm top}\,\frac{(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})}{4(3+2\mathrm{i}\tilde{\nu})(5+4\mathrm{i}\tilde{\nu})(3+4\mathrm{i}\tilde{\nu})},\\
\mathcal{I}^{22}_{\rm top}(-\mathrm{i}\tilde{\nu},\theta_1,\theta_3,\phi_3) &= -\frac{\mathcal{G}_{\rm top}(1+\mathrm{i}\tilde{\nu})}{4(5+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})(5+4\mathrm{i}\tilde{\nu})(3+4\mathrm{i}\tilde{\nu})}\\
&\quad\times\Big\{(2+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})-(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})[c_1^2+c_3^2-3c_1^2c_3^2]\\
&\quad-\frac{1}{2}(1+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad+(2+\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3)\Big\}. \qquad (109)
\end{aligned}$$

The $\eta=-1$ branch follows by complex conjugation. As in the gauge-boson calculation, the $\mathcal{J}_r$ are evaluated by analytic continuation. This extracts the nonanalytic soft coefficient; polynomial contact terms from matching to the hard region are not included.

B.3 Whittaker Seed and Time Integrals

We denote the Whittaker seed $\mathcal{I}$ of Ref. [3] by $\mathcal{K}$. For equal external energies k, internal magnitude $p=rk$, and $u=2r/(1+r)$, the integral seed is defined by

$$\mathcal{K}^{\kappa;p_1,p_2}_{\mathsf{ab}}(u,u)\equiv(-\mathsf{ab})p^{3+p_1+p_2}\int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{p_1}(-\tau_2)^{p_2}e^{\mathsf{i}\mathsf{a}k\tau_1+\mathsf{i}\mathsf{b}k\tau_2}D^{(\kappa)}_{\mathsf{ab}}(p;\tau_1,\tau_2). \qquad (110)$$

In terms of $x_1=-k\tau_1$, $x_2=-k\tau_2$, its time moment is

$$\int_0^\infty\mathrm{d}x_1\mathrm{d}x_2\,x_1^{p_1}x_2^{p_2}e^{-\mathsf{i}\mathsf{a}x_1-\mathsf{i}\mathsf{b}x_2}kD^{(\kappa)}_{\mathsf{ab}}(rk;-x_1/k,-x_2/k)=-\mathsf{ab}\,r^{-3-p_1-p_2}\mathcal{K}^{\kappa;p_1,p_2}_{\mathsf{ab}}(u,u). \qquad (111)$$

At $r=1$, write $\mathcal{K}^{\kappa;p_1,p_2}_{\mathsf{ab}}\equiv\mathcal{K}^{\kappa;p_1,p_2}_{\mathsf{ab}}(1,1)$. The closed expressions are

$$\mathcal{K}^{\kappa;p_1,p_2}_{\pm\mp}=\frac{e^{\mp\mathrm{i}\pi(p_1-p_2)/2+\mathrm{i}\pi\kappa}}{2^{3+p_1+p_2}}\frac{\Gamma[\frac{3}{2}+p_1-\mathrm{i}\tilde{\nu},\frac{3}{2}+p_1+\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2-\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2+\mathrm{i}\tilde{\nu}]}{\Gamma[2+p_1\pm\kappa,2+p_2\mp\kappa]}, \qquad (112)$$

$$\begin{aligned}
\mathcal{K}^{\kappa;p_1,p_2}_{\pm\pm}&=\frac{\mp\mathrm{i}e^{\mp\mathrm{i}\pi(p_1+p_2)/2}}{2^{3+p_1+p_2}}\Gamma[\tfrac{3}{2}+p_1\pm\mathrm{i}\tilde{\nu},\tfrac{3}{2}+p_2\pm\mathrm{i}\tilde{\nu}]\\
&\quad\times\Bigg\{\frac{e^{2\mathrm{i}\pi\kappa}+e^{-2\pi\tilde{\nu}}}{2\pi}\frac{\Gamma[\frac{3}{2}+p_1\mp\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2\mp\mathrm{i}\tilde{\nu},\frac{1}{2}\pm\kappa-\mathrm{i}\tilde{\nu},\frac{1}{2}\pm\kappa+\mathrm{i}\tilde{\nu}]}{\Gamma[2+p_1\pm\kappa,2+p_2\pm\kappa]}\\
&\quad\pm\mathrm{i}\Gamma(3+p_1+p_2){}_3\widetilde{F}_2\left[\begin{matrix}3+p_1+p_2,\ \frac{1}{2}\mp\kappa\pm\mathrm{i}\tilde{\nu},\ 1\\ \frac{5}{2}+p_1\pm\mathrm{i}\tilde{\nu},\ \frac{5}{2}+p_2\pm\mathrm{i}\tilde{\nu}\end{matrix}\middle|1\right]\Bigg\}. \qquad (113)
\end{aligned}$$

Only two scalar time moments are needed to express all three hard coefficients. Define

$$\begin{aligned}
\mathcal{H}_j(q_1,q_2;r)&=\sum_{\mathsf{a},\mathsf{b}=\pm1}\mathsf{ab}\int_0^\infty\mathrm{d}x_1\mathrm{d}x_2\,x_1^{q_1-1}x_2^{q_2-1}(1+\mathsf{i}\mathsf{a}x_1)(1+\mathsf{i}\mathsf{b}x_2)e^{-\mathsf{i}\mathsf{a}x_1-\mathsf{i}\mathsf{b}x_2}\\
&\quad\times F_{j;\mathsf{ab}}(rk;-x_1/k,-x_2/k),\qquad j=0,1. \qquad (114)
\end{aligned}$$

Scale invariance makes the right-hand side independent of k. Expanding the external polynomials with indices $m, n = 0, 1$, put $p_1 = q_1 - 3/2 + m$, $p_2 = q_2 - 3/2 + n$. Integration by parts sends $x_1\partial_{x_1}$ to $-(p_1+1) + \mathsf{i}\mathsf{a}x_1$. Using the definition (110) therefore gives directly

$$\begin{aligned}\mathcal{H}_0(q_1,q_2;r) = &-\frac{\mathrm{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-p_1-p_2}\Big\{(p_1+\tfrac{3}{2}+\mathrm{i}\tilde\nu)[\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1,p_2}-\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1,p_2}]\\ &+\frac{\mathrm{i}(\mathsf{a}-r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1+1,p_2}-\frac{\mathrm{i}(\mathsf{a}+r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1+1,p_2}\Big\},\end{aligned}\tag{115}$$

$$\begin{aligned}\mathcal{H}_1(q_1,q_2;r) = &-\frac{\mathrm{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-p_1-p_2}\Big\{(p_1+\tfrac{3}{2}-\mathrm{i}\tilde\nu)[\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1,p_2}+\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1,p_2}]\\ &-\frac{\mathrm{i}(\mathsf{a}-r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1+1,p_2}-\frac{\mathrm{i}(\mathsf{a}+r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1+1,p_2}\Big\}.\end{aligned}\tag{116}$$

Every seed here has arguments (u,u), and its superscript $\pm$ means $\kappa = \pm 1/2$.

Differentiating the seed definition and integrating the resulting time derivatives by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1,p_2}(u(r),u(r))\Big|_{r=1} = \mathsf{i}\mathsf{a}\,\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1+1,p_2}(1,1)+\mathsf{i}\mathsf{b}\,\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1,p_2+1}(1,1).\tag{117}$$

Both the explicit powers of r and the factors $(\mathsf{a}\pm r)/r$ in (115) must also be differentiated.

Then the coefficients used in the main text are

$$\begin{aligned}\mathcal{P}_{\mathrm{top},1} &= \operatorname*{FP}_{\delta=0}\left[\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)+\frac{2\mathrm{i}}{1+2\mathrm{i}\tilde\nu}\mathcal{H}_1(1+\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)\right],\\ \mathcal{P}_{\mathrm{top},2} &= \operatorname*{FP}_{\delta=0}(\partial_r-1)\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;r)|_{r=1},\end{aligned}\tag{118}$$

The finite part selects the constant term in δ after combining all SK terms and both Whittaker indices.

B.4 Signal Coefficients

With $\rho = 1/2 - \mathrm{i}\tilde\nu$, define the common soft factor

$$\mathcal{B}_{\mathrm{top}}(\tilde\nu) = \frac{2^{1+4\mathrm{i}\tilde\nu}\Gamma(\frac{1}{2}-\mathrm{i}\tilde\nu)^4 f_0(\rho,\rho)}{32\pi^2(4\pi)^{3/2}(\rho-3)(\rho-2)(4\rho-7)(4\rho-5)}.\tag{119}$$

The factor includes the normalization of the two soft lines, the spin trace and the four orderings, together with the phase convention in (58): $8c_+^2\,16^{2\mathrm{i}\tilde\nu} = 2^{1+4\mathrm{i}\tilde\nu}\Gamma(1/2-\mathrm{i}\tilde\nu)^4/\pi^2$. Substituting (109) gives

$$\begin{aligned}A_{1,\mathrm{top}} &= \mathcal{B}_{\mathrm{top}}(\rho-1)(2\rho-3)\left[4(\rho-3)(3\mathcal{P}_{\mathrm{top},1}^2+2\mathcal{P}_{\mathrm{top},1}\mathcal{P}_{\mathrm{top},2})+(2\rho-5)\mathcal{P}_{\mathrm{top},2}^2\right],\\ A_{2,\mathrm{top}} &= -\mathcal{B}_{\mathrm{top}}\mathcal{P}_{\mathrm{top},2}^2(\rho-1)(2\rho-3)^2, \qquad A_{3,\mathrm{top}} = -3A_{2,\mathrm{top}},\\ B_{\mathrm{top}} &= -2\mathcal{B}_{\mathrm{top}}\mathcal{P}_{\mathrm{top},2}^2(\rho-1)(\rho-2)(2\rho-3),\\ D_{\mathrm{top}} &= \mathcal{B}_{\mathrm{top}}\mathcal{P}_{\mathrm{top},2}^2(\rho-2)(2\rho-5)(2\rho-3).\end{aligned}\tag{120}$$

All couplings and the color factor remain outside these coefficients, as specified in (57).

For the opposite soft branches, the leading endpoint matrices are $L_{\mathrm{L}}^{-} = \mathbf{1}_2$, $L_{\overline{\mathrm{L}}}^{-} = -\boldsymbol{\epsilon}$, $U_{\mathrm{L}}^{-} = \boldsymbol{\epsilon}$ and $U_{\overline{\mathrm{L}}}^{-} = -\mathbf{1}_2$, with scalar prefactor $+c_-/2$. The vertex-type sums are

$$V_L^{-+,(0)} = F_{1;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\mathbf{1}_2, \qquad V_R^{+-,(0)} = F_{1;\mathsf{a}_4\mathsf{a}_3}(k_R;\tau_4,\tau_3)\mathbf{1}_2. \tag{121}$$

The superscripts here refer to the soft mass branches. The soft prefactor is $-|c_+|^2(q^2)^{-\mathrm{i}\tilde{\nu}}(K^2)^{\mathrm{i}\tilde{\nu}}$; its minus sign cancels the loop sign. There are two branch assignments and two external orderings at each end. Together with the Pauli trace these supply a factor 16. Using $|c_+|^2 = 1/[4\cosh^2(\pi\tilde{\nu})]$ and (**??**) gives the coefficient $2\tilde{\nu}(1+\tilde{\nu}^2)/[3\pi\sinh(2\pi\tilde{\nu})]$ in (**??**).

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